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IdeaBlocks: Expressing and Reusing Divergent Intents for Graphic Design Exploration using Generative AI

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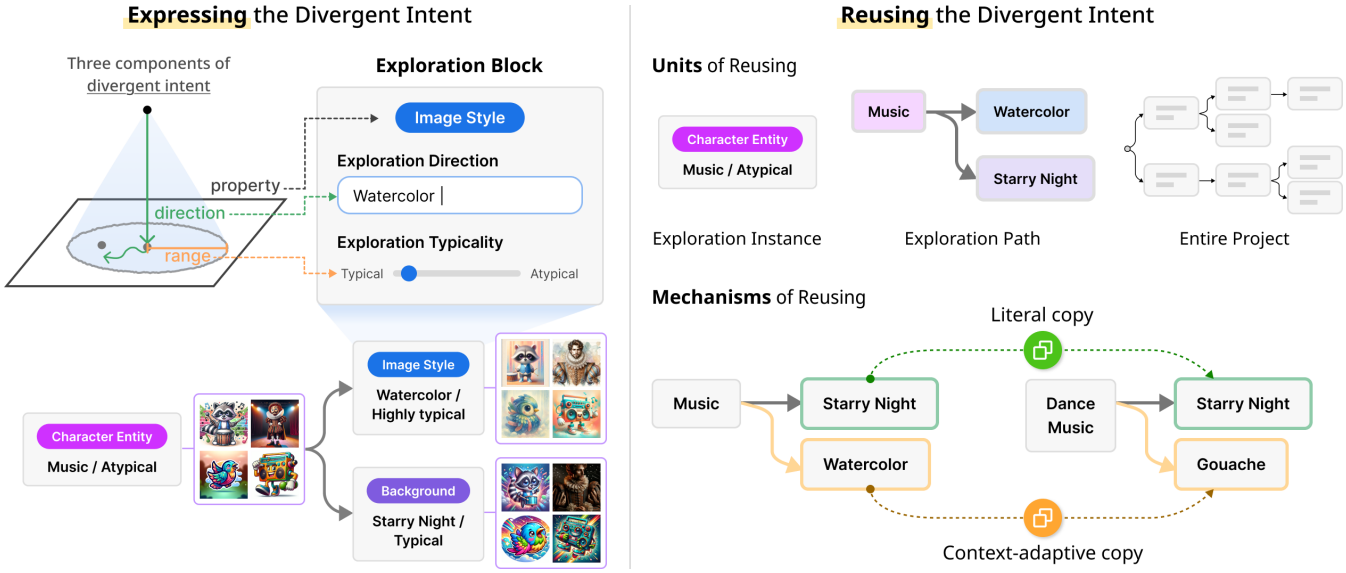


Figure 1: Overview of IdeaBlocks. (Left) Users' divergent intents can be modularized into exploration blocks with three components: property, direction, and range. These blocks can be chained across multiple properties. (Right) Previously created blocks, paths, or entire projects can be reused via literal copy (preserving the original intent) or context-adaptive copy (adapting it to a new context), enabling efficient and iterative exploration.

Abstract

While designers increasingly leverage Generative AI for divergent exploration, current interaction is optimized for convergent refinement, forcing users to specify fixed targets rather than open-ended search spaces. Based on a formative study (N=7), we define the anatomy of *Divergent Intent*, comprising property, direction, and range, and identified two critical barriers: the lack of mechanisms to explicitly shape the parametric boundaries of exploration and the difficulty of reusing successful search strategies. We present IdeaBlocks, where users can modularize divergent intents into *Exploration Blocks*. Users can reuse prior intents at multiple levels (block, path, and project) with options for literal or context-adaptive reuse. In our comparative study (N=12), participants using IdeaBlocks explored 2.13 times more images with 12.5% greater visual diversity than the baseline, demonstrating how structured intent expression and reuse support divergent exploration. A three-day longitudinal study (N=6) further revealed how different reuse mechanisms allowed distinct creative strategies, offering design implications for future intent-aware design support tools.

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CCS Concepts

• **Human-centered computing → Interactive systems and tools**.

Keywords

Creativity support tool, Design exploration, Generative AI, Graphic design, Divergent intent

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1 Introduction

Design exploration is essential to the creative process, such as graphic design [19], and is characterized by the iteration of divergence and convergence [3, 20]. In diverging phases, designers operate with ***divergent intent*** to experiment widely within a design *space* to *"get the right design"*, while in convergent phases, they adopt *convergent intent* to narrow toward a specific *point* and precisely represent that fixed design idea to validate and *"get the design right"* [8]. Divergent intent is not a fixed design goal, but a tentative and revisable orientation for exploration that guides how designers probe and adjust the space of possibilities through interaction with emerging outcomes [17, 56].

Generative AI presents a unique opportunity to support divergent intent in graphic design exploration [35]. By enabling rapid generation of visuals from text, it aligns with early-stage prototyping, where low-cost artifacts are used to explore a broad design space [40]. However, a structural mismatch remains between this potential and current interaction models. Existing mechanisms are largely optimized for convergent intent, encouraging users to specify a target outcome through prompt refinement [6, 69] or reference images [57]. This prematurely narrows exploration toward a single point in the design space, which is problematic when designers' goals are still forming. Recent systems attempt to broaden exploration by suggesting diverse alternatives [26, 62]. Yet users still lack mechanisms to *control* their divergence, as they lack parametric means to define the scope and boundaries of their exploration, which are continuous and evolve as designers' intents develop.

To probe this gap, we conducted a formative study with 7 participants, focusing on the challenges of articulating divergent intents when exploring with generative models. From our observation, we identified the anatomy of divergent intent comprising three components—*property*, *direction*, and *range*—which together define the *space* a user intends to explore. However, current generative AI interactions lack mechanisms to express these dimensions, often leading to premature convergence or unintended results. We further found that once designers discover a satisfactory exploration strategy, it becomes a reusable unit of design knowledge. Rather than reusing individual prompts or outputs, designers seek to reuse the *structure* of exploration—the underlying logic that defines the search space. This suggests that divergent intent reuse operates at a higher level, capturing parametric exploration logic rather than single instantiated prompts.

We propose IdeaBlocks, a system that enables users to structure their divergent intents during early-stage exploration into modular units called *Exploration Blocks*, based on three components of *property*, *direction*, and *range*. For instance, a designer creating a mascot character might create a block for *Character Entity*, set the direction to "Music," and adjust the typicality slider toward *Atypical* to explore more divergent variations. Then, they can chain it to other blocks for different design *properties* such as *Image Style* to explore multiple dimensions of their divergent intent simultaneously. IdeaBlocks also supports reusing previous divergent intents in three units: block-level (individual exploration instance), path-level (sequence of blocks), and project-level (entire exploration process). At each level, users can choose to reuse intents exactly as before or adaptively in a new exploration context.

We conducted a within-subjects comparative study with 12 designers to evaluate how modularizing and reusing divergent intents in IdeaBlocks supports divergent design exploration. The results demonstrate that IdeaBlocks enhanced participants' ideational fluency and the diversity of their exploration outcomes. They tried 1.77 times more inputs to the model, facilitated by the structured expression of their divergent intent, resulting in 2.13 times more generated images explored within the same time. Their image generation rate increased over time, indicating more sustained exploration driven by active reuse of previous intents. They also explored 12.5% more visually diverse outcomes, measured by CLIP-based embedding distances across the generated images. Linkography analysis [23] revealed that their explorations were more structured and interconnected with higher entropy, indicating IdeaBlocks supports broader, more structurally diverse, and less linear idea development.

To further investigate how intent-reuse features of IdeaBlocks shape design exploration in practice, we conducted a three-day longitudinal study with 6 designers. For block-level reuse, literal reuse of visual properties such as image styles acted as "personal palettes", while semantic properties such as entities or poses were more often adapted to context. Path-level reuse helped participants recognize their own recurring exploration strategies and reuse them as template-like scaffolds. At the project level, reuse served two purposes: bootstrapping (to avoid blank starts by building on small, understandable projects) and broadening options (drawing inspiration from others' intermediate strategies). From these observations, we suggest design implications for future systems to better incorporate intent-reuse features for creative exploration.

Our contributions are threefold:

- Formative study findings providing an understanding of the limitations of current generative AI in supporting divergent intent, and a conceptual framework that decomposes divergent intent into three components: *property*, *direction*, *range*.
- IdeaBlocks, which externalizes divergent intents into manipulable and reusable units, demonstrated to be effective through a comparative study.
- Design implications for supporting intent reuse in design exploration, derived from a longitudinal study that reveals how and why reuse features are employed across different contexts.

2 Related Work

We review related research on (1) generative AI tools that support design exploration and (2) systems that model and reuse user intent in generative workflows.

2.1 Generative AI Tools for Design Exploration

2.1.1 Supporting Divergent Exploration via Generation and Scaffolding. One strategy to solve ill-defined design problems is to generate diverse variations to expand the solution space [59]. Research has shown that exploring these variations *in parallel*, rather than serially refining a single concept, leads to more divergent outcomes and reduces fixation [18, 65]. Tools such as Gem-Ni [73] and Mixplorer [32] automate this by creating parallel variations to broaden potential outcomes. Similarly, Koyama et al. [38] asynchronously suggest unexplored design candidates using Bayesian optimization. Another approach facilitates divergence by synthesizing novel outputs through the recombination of distinct sources, as seen in CreativeConnect [12], PopBlends [67], and Artinter [14]. Dreamsheets [2] utilized spreadsheet-based prompt permutations to generate volumes of variations.

Another approach is to help users widen the design space to overcome fixation [18]. Systems like Luminate [62], MetaMap [30], and Dream Lens [45] help users navigate large generated design spaces — through hierarchical taxonomies, semantic attributes, or interactive visualization of large-scale design datasets. Similarly, POET [26] and DesignAID [9] utilize LLMs to reason about the problem space and suggest diverse design directions or keywords that users might not have considered.

While these systems successfully broaden the options to foster divergent thinking, they rely on selection-based interactions, where users choose from a broadened taxonomy or list of generated candidates. In contrast, IdeaBlocks complements this approach by giving users direct control over the direction and range of variation, enabling them to actively shape the exploration space itself.

2.1.2 Managing and Structuring the Exploration Process. Design ideation is non-linear and often involves branching, merging, and revisiting ideas. Users' exploration strategies vary depending on the maturity of their ideas [70]. Therefore, tools have been helping users manage and navigate their exploration process. One line of work is to use moodboards [10, 19], as used in the systems like Funky Wall [42], SemanticCollage [37], May AI [36], DesignPrompt [47], and MoodCubes [28]. Beyond moodboards, Charrette [46] supports designers in curating iteration histories and navigating previous design decisions alongside meeting notes. More recent attempts are to use a graph-based interface to represent those exploration paths. Expandora [13] incorporated a mindmap interface to navigate prompt-generated ideas, and Graphologue [29] and Sensecape [63] visualize LLM interactions as node-link diagrams. Other systems, including DeckFlow [16], XCreation [71], Spellburst [4], and OptiMuse [75], adopt graph-based interfaces that let users express, organize, and reflect on ideation paths through spatial structuring and iteration.

Supporting non-linear workflows aligns closely with the iterative nature of the creative process. Inspired by this, IdeaBlocks was also designed as a canvas-based node-link system to allow users to interact with their modularized intents in a non-linear manner.

2.2 Intent Modeling and Reusing in Creative Systems

As generative AI becomes more integrated into creative workflows, designers often express intentions through text prompts. However, text modality can often be limiting, especially in open-ended ideation where intentions are vague or evolve dynamically. Recent studies investigate these struggles [44, 74], highlighting the need for systems that better support the expression, resolution, and reuse of creative intent.

2.2.1 Expressing and Resolving Creative Intents. Recent research has introduced systems to help users express creative intent more effectively for generative models. One major approach focuses on prompt refinement, where systems like Promptify [6], RePrompt [68], or PromptCharm [69] utilize visualizations or multimodal prompt engineering to iteratively guide users toward a prompt that accurately reflects their mental image. Similarly, PromptMap [1] and Patchview [15] provide structured alternatives for prompts to make them more interpretable. Another approach provides structured input instead of a text prompt. For example, GenQuery [61] and Opal [41] guide structured searches of visual concepts. DesignPrompt [47] and StyleFactory [76] incorporate multimodal inputs to capture users' aesthetic preferences.

However, in design exploration settings, users often begin with open-ended ideas with high uncertainty regarding their goals [58, 64], inevitably leading to the use of vague prompts. To address this, a recent approach incorporates proactive interaction to resolve ambiguity. Hahn et al. [25] propose methods to visualize the model's understanding and ask clarification questions to specify user intent. CARE [48] employs a multi-agent framework for personalized query refinement, and Stylette [33] interprets high-level styling goals into specific low-level CSS properties.

While prior work effectively helps articulate prompts, its goal is to reduce ambiguity and guide generation toward a specific outcome, which fundamentally supports convergent intent. In contrast, IdeaBlocks embraces the inherent uncertainty of early-stage ideation not as a problem to be solved, but as a parameter to be controlled. Rather than attempting to fix the vague intent, IdeaBlocks allowed users to explicitly express divergent intent by defining the direction and range of variance.

2.2.2 Reifying and Reusing Creative Intents. Another line of research explores how users' intents can be retained and reused. Drawing from the instrumental interaction model [5], recent frameworks have proposed reifying interactions with AI into manipulable objects [34, 43]. AI-Instruments [54] and Memolet [72] turned user prompts or chat interactions into reusable widgets or memory objects to enhance operational efficiency. IntentTagger [22] introduces micro-prompts for granular specification that can be reused during the co-creation process. Not only for the piece of text prompt, but the recent systems, such as Brickify [57], create persistent design tokens from images to facilitate the reuse of specific visual attributes.

Prior approaches largely focus on reusing static content, operational commands, or local specifications. Motif [31] reuses expert creative patterns to scaffold novice work, and SelPh [39] progressively learns users' personal preferences from editing history to

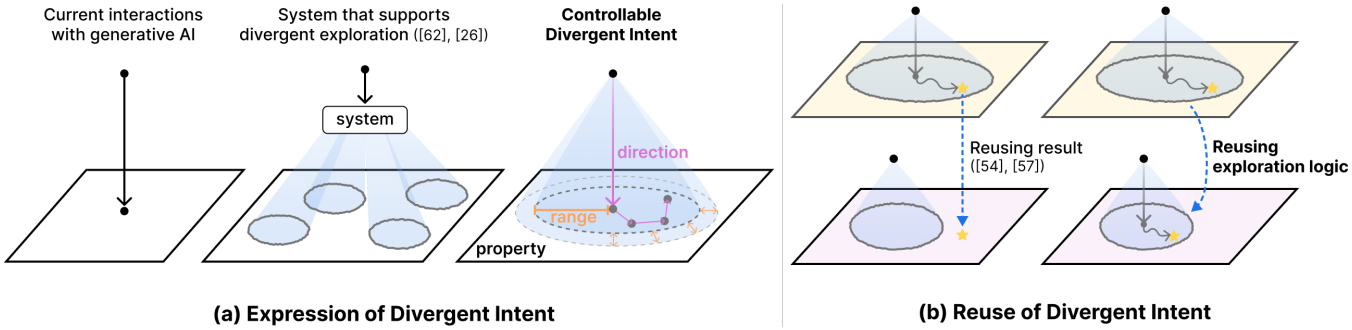


Figure 2: Conceptual distinction between IdeaBlocks and prior works. (a) Expression: Current GenAI interactions are optimized for specifying a single point. Prior research adds divergence but lacks user control over its boundary. IdeaBlocks enables users to shape parametric boundaries (Property, Direction, Range), freely navigating a property space and controlling the range of variation. (b) Reuse: Prior methods reuse static assets or prompts. IdeaBlocks reuses the exploration vector (Direction + Range), decoupling the strategy from the content to transfer successful exploration logic across design contexts.

guide future actions. More recently, Brickify [57] reuses specific visual properties to ensure consistency, while AI-Instruments [54] reuses functional controls for efficiency, and both primarily support convergent objectives. In contrast, IdeaBlocks reifies the exploration logic by modularizing the strategy for probing the design space, allowing designers to transfer their divergent intent across different contexts rather than simply repeating fixed actions.

3 Formative Study

To better understand the limitations of using current generative AI tools for early-stage design exploration, we conducted a formative study, especially focused on the challenges when expressing and evolving their divergent intent. We recruited 7 participants (6 females; age M = 24.86 and SD = 3.34) through an online recruitment post. All participants had prior experience with image generation models for design tasks and had worked on at least three graphic design projects before. Details about the participants are in Appendix A.1. The study lasted for an hour, and participants were compensated 25,000 KRW (approx $17 USD).

3.1 Study Procedure

Participants were asked to explore diverse mascot character designs for their company (for practitioners) or department (for students) with image generation models for 40 minutes. This task was chosen for its open-ended nature, which encouraged divergent thinking, while also being sufficiently familiar to all participants with different design experiences. Following the session, we conducted a 15-minute semi-structured interview.

We provided a custom image-generation interface to facilitate think-alouds about their intent during exploration. The system supported four methods for image generation: (1) text-to-image prompting, (2) image-to-image generation with text guidance, (3) text-guided inpainting, and (4) generating variations from an image. We used Stable Diffusion XL [50] for text-to-image, InstructPix2Pix [7] for image-to-image, and DALL·E 2 API [1] for inpainting and variations. Whenever participants use one of them, the interface displays a pop-up message asking them to articulate their current intents and expectations. After reviewing the generated results, the tool prompted them to reflect on whether the results accurately reflected their intention, and if not, what they thought the reason was. In total, we collected 158 data points, each consisting of a participant's expressed intent, system action, and participant's assessment of how well the outputs aligned with their intent.

We conducted an inductive thematic analysis of the collected data and interview data. Open coding was performed iteratively after each session until thematic saturation was reached.

3.2 Findings

3.2.1 Characterizing Divergent Intents in Design Exploration. Design exploration begins with an open-ended intent to explore the *space* of possibilities, rather than a single *point*. We observed that participants rarely articulated such intent as a fixed design goal, but in terms of how they wanted to orient exploration by specifying where to look, what to vary, and how broadly to explore, while remaining open to revision. This suggests that intent in early-stage exploration functions as a tentative, revisable probe into the design space [17, 56], which we define here as *divergent intent*.

Through our thematic analysis of participants' articulation attempts of this exploration space, we identified that *divergent intent* could be structured with three components (Figure 2). While participants did not use these terms explicitly, their articulations consistently reflected attempts to specify what aspect to vary, in which direction, and how broadly — which we abstract into *property*, *direction*, and *range*, respectively (see Appendix A.2 for the full codebook, including representative quotes and examples from participants).

- *Property*: Dimension; *What* aspect to explore (e.g., entity of a mascot, background setting, image style)
- *Direction*: Center; *Where* to center the exploration within the property (e.g., "a cat mascot" for entity, "watercolor style" for image style)
- *Range*: Magnitude; *How far* to deviate from that center (the breadth of variation of the results)

[1] https://openai.com/index/dall-e-2/

3.2.2 Mismatch Between Divergent Intent and Convergent Controls.

Lack of Control over Exploration Range. As participants initiated their explorations, they frequently shifted between seeking results closely aligned with their original input (convergent) and pursuing options that ventured into atypical territory (divergent). Most prominent challenge was their inability to control this *range* of exploration. Current interfaces do lack explicit controls for the desired breadth of divergence, and to work around this, participants tried to push their prompts with vague or extreme modifiers like "various styles" or "doing something" in their prompts (P2, P3) or by relying on the variation feature (P1, P3, P5-7). However, these strategies were unreliable, often overshooting into randomness or undershooting into homogeneity, because the system could not interpret the intended magnitude of variance. While recent approaches [26, 62] have attempted to broaden exploration support by structurally suggesting various design dimensions, they primarily support *categorical* breadth rather than *parametric* variance, preventing users from manipulating the magnitude of divergence along those dimensions (Figure 2). Therefore, we need a way to operationalize divergence not as a discrete state, but as a continuous and tunable parameter, allowing users to explicitly calibrate the scope of exploration to match their evolving intent.

Friction in Evolutionary Steering. As exploration unfolds, users' design intentions on what to explore next naturally evolve in response to the results they see [17]. Reflecting this, all participants repeatedly refined their prompts after each iteration, yet this process introduced substantial friction in two key ways. First, users had to mentally translate the visual preferences they observed back into explicit text descriptions. Second, as the text prompt entangles all aspects of the image, modifying the prompt to refine the *direction* of a specific *property* often unintentionally affects other *properties*. Across sessions, participants expressed a need for **bounded iteration**.

Several participants (P1, P2, P5, P7) envisioned that when starting from a vague direction, the system could suggest candidate directions within the property and range that they're interested in, allowing them to refine the exploration by accepting or rejecting these options. This highlights the need for a lightweight mechanism to "lock" the exploration scope and iteratively steer the direction within it.

3.2.3 Lack of Support for Reusing Past Divergent Intents.

Once participants found an exploration trajectory that suited their evolving intents, they looked for ways to reuse that same exploration pattern to new contexts, applying the underlying logic of how they probed the design space, rather than merely reusing the results they liked.

Some participants tried to do this by copying and pasting descriptions from the previous prompts (P1, P2, P3). However, participants found that simply reusing parts of a text prompt failed to transfer this logic, as text prompts entangle the content (what to draw) with the exploration parameters (how to vary it). For instance, P1 initially explored "drawing of a soft cloud" and appreciated that the generated results helped her see a broad range of diverse drawing styles. Hoping to apply this same diversity of style to a specific character, she reused the phrase "soft drawing style" in a new prompt: "young boy with his head and legs sticking out of a cloud, soft drawing style." However, instead of producing the diverse range she liked, the model converged on a homogenous set of typical pencil drawings. Because the user could not explicitly decouple and reuse the strategy of divergence from the text prompt, they lost the exploration space they had found and wanted to reuse.

We also observed that the scope of reusing previous intent varied across situations. For example, when participants wanted to reuse divergent intents tied to their global preferences, they wanted to reuse them across different projects. When the intents were tied to more specific ideas, they wanted reuse to remain confined to that idea and to be able to escape it when exploring other ideas. We conceptualize this variation along two dimensions: *unit*, or how much of the previous exploration is reused, and *adaptivity*, or how fixed versus flexible that reuse should be across different exploration contexts. These two axes form the basis of how IdeaBlocks supports the reuse of divergent intent.

3.3 Design Goals

Based on the findings, we derived two design goals (DGs) for the system to address the challenges faced by designers during the use of the generative models in early-stage design exploration:

DG1. Support expressive articulation of divergent intents. Designers struggled both to articulate the direction and the range of the divergent intent. The system should provide mechanisms for expressing direction of divergent intents more flexibly, and allow explicit control over how broadly to explore around a given direction.

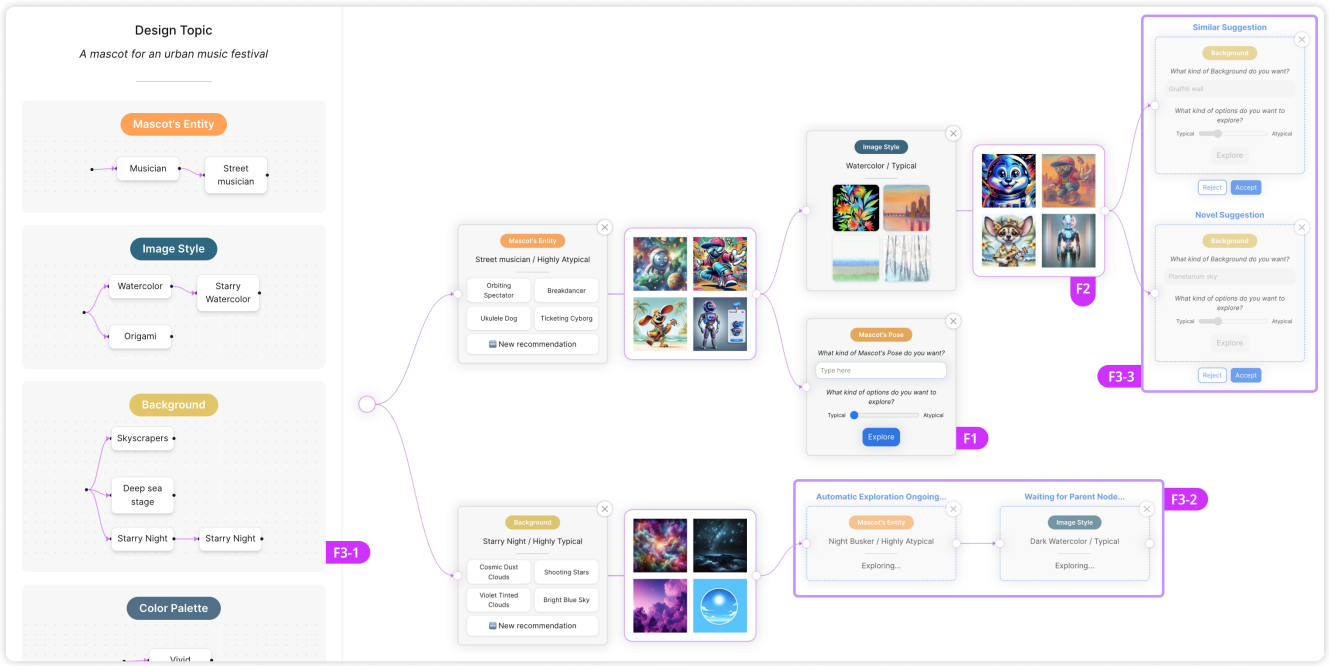
DG2. Enable flexible reuse of past divergent intents. Designers often wanted to reuse the aspects of their explorations. Their reuse needs varied across scopes. A system should externalize intents into reusable units that can be retrieved and reused in diverse ways.

4 IdeaBlocks

Based on the design goals, we implemented IdeaBlocks to support designers' early-stage exploration by enabling the expression and reuse of divergent intents. The core of IdeaBlocks is the concept of an ***Exploration Block***—a modular unit that captures the user's divergent intent. Each block (Figure 4) includes three components identified in our formative study: property (defined by selecting a type of exploration block), direction (specified via a keyword input), and range (controlled through the typicality slider).

To ease the challenge of steering directions, IdeaBlocks augments user input with system-generated textual and visual suggestions, enabling users to refine directions by selecting, modifying, or extending examples. To control the range, we intentionally chose *typicality* as a parameter because it captures an implicit but common concern in early-stage exploration — how far to deviate from a familiar center — and is easy for designers to reason about across different design properties. Typicality serves as one concrete instantiation of how the range of divergent intent can be externalized and controlled. We discuss how this approach can be expanded in Section 8.1.

Through this structure, IdeaBlocks enables users to explicitly define their divergent intent and modularize it into *Exploration*



(a) Blocks Storage Panel (b) Exploration Canvas

Figure 3: Screenshot of the IdeaBlocks system with (a) the *Blocks Storage Panel* and (b) the *Exploration Canvas*. Users express their divergent intent through structured inputs in *Exploration Blocks* (F1), which can be chained to build on prior ideas across different properties. Users can reuse individual blocks (F2-1), entire exploration paths (F2-2), or the whole project, with optional adaptation.

Blocks (DG1). These blocks can be chained to preserve prior exploration and build new explorations incrementally. Previously explored blocks and paths are stored as a reusable format, allowing users to efficiently revisit and reuse their past explorations (DG2).

4.1 Interface Walkthrough & Features

IdeaBlocks provides a canvas interface for creating and interacting with *Exploration Blocks* (Figure 3). In this section, we illustrate its key features through a scenario with John, a concept artist designing a mascot character for a music festival.

4.1.1 Generate a List of Properties. The exploration begins by entering a design topic, and IdeaBlocks uses GPT-4o to suggest relevant design properties, which are shown in the left panel (Figure 3 (a)). Two types of properties are generated, either text-based (e.g., "Character's Entity") or image-based (e.g., "Image Style"). Users can also add their own properties to explore.

> John wants to design a mascot for the music festival organized by his team, and he inputs the topic into IdeaBlocks. The system presents a list of properties to explore, including "Mascot's Entity", "Image Style", and "Background".

4.1.2 Expressing Divergent Intent into Exploration Block. For each design property, users create an *Exploration Block* by entering a keyword and adjusting a typicality slider. The keyword sets the initial direction of exploration (e.g., 'Street musician'), and the typicality level (1-5) controls the exploration breadth: lower typicality levels generate more aligned and conventional results, while higher levels produce more divergent outputs.

Based on these inputs, the system generates four suggestions, shown as text or images depending on the property type. Technical details for generating suggestions are presented in Section 4.2.1. Users can delete undesired suggestions or request similar or more distant suggestions, and the system updates exploration direction based on each user action. The images generated from each suggestion appear in a *Result Block* next to its corresponding *Exploration Block*.

Each *Exploration Block*, along with its parameters and outcomes, is stored and summarized in the left panel. These stored blocks can later be reused for further exploration, as described in Section 4.1.4.

> John begins by exploring "Mascot's Entity" and creates an *Exploration Block* for this property. He enters 'Street Musician' as the initial direction and sets the typicality level to 5 to explore broadly and discover unexpected ideas. The system generates suggestions such as 'Ukulele Dog' and 'Ticketing Robot' (left side of Figure 4). John refines the direction by removing irrelevant ones and getting more suggestions around preferred ones. He then reviews the images generated in the *Result Block*.

In addition to manually entering a keyword, users can receive system-generated direction recommendations, which draw on the user's prior exploration history and the current context (i.e., connected blocks and property type). Using GPT-4o, the system provides both familiar and novel directions.

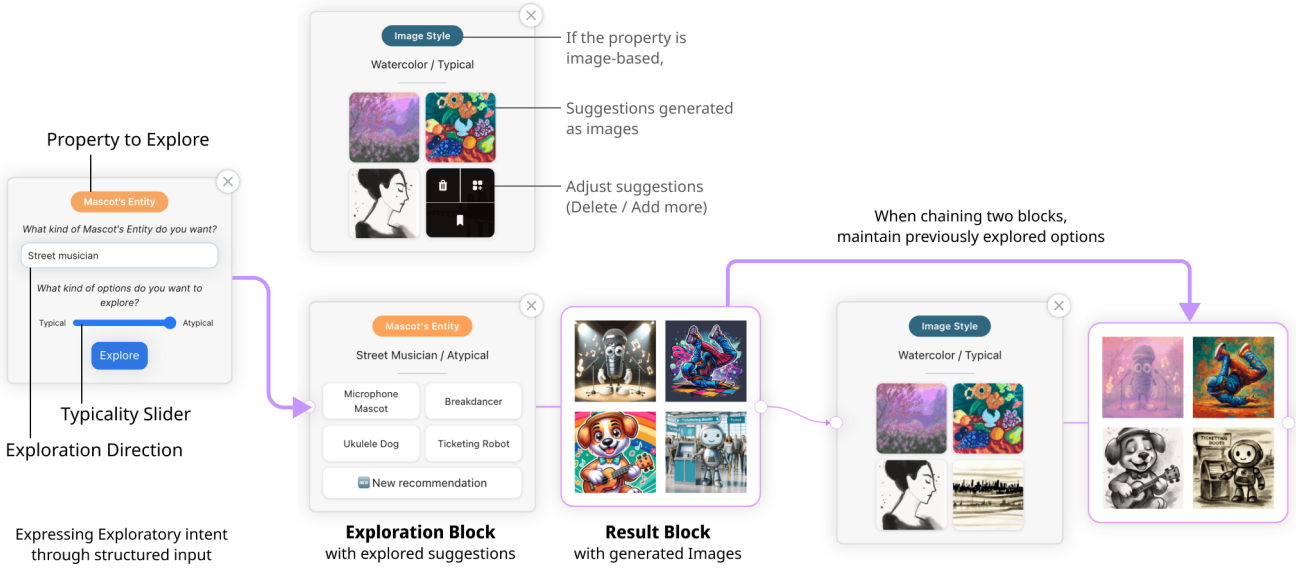


Figure 4: An *Exploration Block* comprises a property type, exploration direction, and typicality slider. Based on these settings, IdeaBlocks generates four suggestions as text or images, depending on the property (e.g., text for Mascot's Entity, images for Image Style), which users can delete or request more of. The adjacent *Result Block* displays generated images. When chaining blocks, each *Result Block* carries forward previously explored options, allowing users to build on prior intent across properties.

4.1.3 Chaining Multiple Blocks. Users can chain multiple *Exploration Blocks* to build on earlier ideas across different properties. When a new block is linked to an existing one, its setting, suggestions, and generated results become input for the next stage of exploration, and images generated in subsequent *Result Blocks* reflect both the current intent and the accumulated exploration history. Technical details of this image generation process are in Appendix B.1.

> After exploring different character entities, John moves on to the next property, "Image Style". He creates a new *Exploration Block* and connects it to the previous one. He explores typical watercolor styles and adjusts the system's suggestions, just as he did with the "Mascot's Entity". The generated images shown in the *Result Block* reflect not only the watercolor style, but also incorporate the previously explored character entities from the connected block (right side of Figure 4).

4.1.4 Reusing Previous Intentions. To support efficient and flexible exploration, IdeaBlocks provides multiple features for reusing earlier divergent intents along two complementary axes: the unit of reuse (block, path, or project), and the level of adaptation (literal vs. adaptive).

Block-level Reuse. At the most granular level, users can reuse individual *Exploration Blocks*, which corresponds to each exploration instance. All previously created blocks across projects are stored in the left sidebar (Figure 5 (a)). As users explore, IdeaBlocks uses GPT-4o-mini to generate a graph in the left panel that visualizes how their exploration directions evolve for each property. Selecting a block from this graph places a new copy onto the canvas, which users can then connect to new exploration paths. IdeaBlocks supports both literal reuse (reusing the block as-is with the same parameters and suggestions), and adaptive reuse (generating adjusted variants of the block based on the current exploration context). All adaptations (at the block, path, and project levels) are generated using GPT-4o with the same prompts.

> During the exploration, John explores a new background setting, 'Deep sea stage', and wants to reuse a previously explored 'Watercolor' image style on it. He selects the saved block in the sidebar and connects the copied block to the new setting. IdeaBlocks then presents two options: a literal version reproducing the earlier style suggestions, and an adaptive version retaining the watercolor aesthetic while tailoring it to the 'deep sea' context.

Path-level Reuse. Users can also reuse an entire sequence of previously crafted blocks by copying and pasting an exploration path (Figure 5 (b)). By choosing a path to reuse and a point on the canvas to paste, two options are given: (1) a literal copy that replicates all blocks as-is, or (2) a context-adaptive copy that adjusts block directions and typicality levels based on the context of the target point. Once pasted, the system creates blocks with appropriate parameters and automatically continues the exploration, which enables users to efficiently branch off from a new starting point while preserving a familiar exploration trajectory.

> Later, John recalls a successful exploration path about 'Street Musician' character paired with a 'Watercolor' style. Eager to reuse it, he selects that path and pastes it on the new 'Skyscraper' background setting. IdeaBlocks offers literal copy and three different context-adaptive variants. In one adaptive version, the character is reimagined as a 'Night Busker', and the style shifts to 'Gouache' to better align with the current 'Skyscraper' theme. John selects this version, quickly branching into a new yet familiar sequence of exploration.

Project-level Reuse. At the broadest unit, IdeaBlocks allows users to import an entire project, including all paths and blocks.

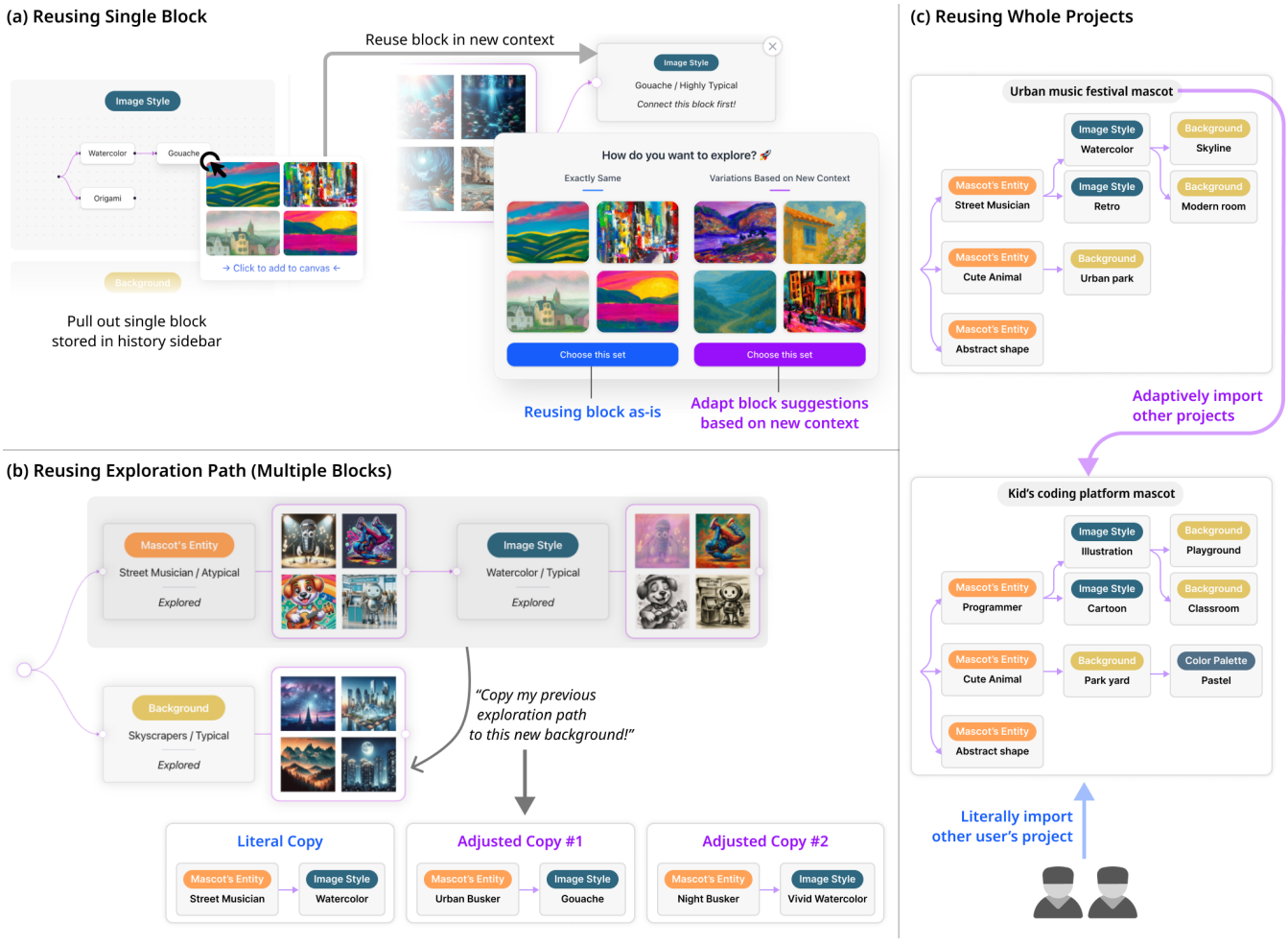


Figure 5: IdeaBlocks's four features for reusing prior exploration. (a) A previously created *Exploration Block* can be pulled from the history sidebar and placed on the canvas as-is (literal) or with suggestions adapted to the new context. (b) An entire *Exploration Path* can be copied as a literal duplicate or as context-adaptive variants with adjusted block parameters. (c) A previous project can be adaptively imported to bootstrap a new topic, or another user's project can be literally imported to borrow their exploration structure.

Users can reuse their own or other users' projects on similar topics (Figure 5 (c)). As with block- and path-level reuse, IdeaBlocks offers literal reuse (preserving all blocks as-is) and adaptive reuse, which adjusts the parameters of each block to better align with the new project topic. Imported projects appear on the canvas, which users can then extend, branch from, or merge with their current exploration.

> After working on the mascot design for a 'Music Festival', John starts a new mascot design for 'Kids' coding platform'. Rather than starting from scratch, he imports the earlier project about a music festival mascot. IdeaBlocks offers literal and adaptive reuse options; in one adaptive version, John found that the 'Street Musician' entity is reimagined into 'Digital Nomad', and 'Watercolor' style shifts to 'Illustration' to better suit a playful, child-friendly theme. By choosing this, John can quickly bootstrap his new exploration. John can also browse projects created by other users working on similar topics. While looking at others' explorations, he discovers an interesting branch that combines an 'Animal Character' with 'Sticker' style. He imports this project and borrows that segment into his own flow. This allows him to blend familiar ideas with fresh perspectives from others' explorations.

4.2 Technical Details & Pipeline Overview

IdeaBlocks was built as a web-based system using ReactJS[2] and React Flow[3]. IdeaBlocks includes a pipeline that operationalizes divergent intent by translating user-specified intent parameters into system-generated suggestions. The full prompts used for each step of this pipeline are provided in the supplementary material.

[2]https://react.dev/
[3]https://reactflow.dev/

4.2.1 Divergent Intent into System-Generated Suggestions. A key mechanism of IdeaBlocks enables users to explicitly define the design space by controlling three parameters of divergent intent: property, direction, and range. We designed and evaluated one concrete instantiation of this concept through our pipeline that translates users' input on visual property, direction keyword, and typicality range into controlled visual or textual suggestions.

Pipeline Design. IdeaBlocks first generates a diverse pool of candidate suggestions using a multi-step prompting strategy. Given the input property and direction, GPT-4o generates ten alternative directions spanning from literal extensions of the input to more distant interpretations. For each alternative direction, the system produces multiple suggestions, yielding a total of 100 text-based or 50 image-based candidates. This pool serves as the raw material for the subsequent algorithmic filtering and selection steps.

Next, to follow the user-specified 'range' (typicality level), the system computationally filters the candidate pool. We define typicality as how easily a suggestion can be mentally evoked from the user's input direction (i.e., how commonly the two co-occur conceptually or linguistically). To operationalize this, we measure co-occurrence-based similarity between the input direction and each candidate: for text, using GloVe [21, 49]; and for images, using the CLIP score [52]. By ranking and filtering candidates based on these calculated metrics, the system ensures that the output aligns with the user's intended magnitude of divergence.

Finally, to support iterative refinement of 'Direction', the system employs K-means clustering to structure them into groups. We cluster the candidates using Sentence-BERT [53] (text) or ViT-B/32 CLIP embeddings [52] (image) and present the centroids of the four most distinct clusters first. When a user selects or refines one of these suggestions, the system utilizes the embedding of the chosen centroid as the new semantic anchor, dynamically shifting the direction.

Pipeline Evaluation. We evaluated the pipeline to examine whether its typicality-based filtering aligns with human judgment. We constructed an evaluation set using five $(Property, Direction)$ pairs from the formative study (3 text-based, 2 image-based). Using these pairs, the pipeline generated 100 candidates per text property and 50 per image property, totaling 400 suggestions. After that, we randomly selected 100 suggestions (60 text-based, 40 image-based) as a final evaluation set. We recruited 3 raters with prior design experience and asked them to label typicality levels for each system-generated suggestion given the input direction and property. Inter-rater reliability (Krippendorff's Alpha) was 0.75 for text-based and 0.69 for image-based suggestions, indicating substantial agreement. Final labels were established by averaging their ratings.

We then examined the alignment between human-labeled typicality levels and the pipeline's typicality ordering. For text-based suggestions, the system's scores showed moderately strong alignment with human annotations (Pearson r=0.68, Spearman ρ=0.67). For image-based suggestions, correlations were moderately aligned (Pearson r=0.59, Spearman ρ=0.58). Together, these results indicate that the pipeline produces suggestions whose typicality ordering broadly corresponds to human perception.

We note that the metrics that we chose serve as proxies for typicality: GloVe captures thematic co-occurrence rather than categorical typicality (i.e., how representative an instance is of its category [55]), and CLIP measures image-text correspondence rather than concept-level co-occurrence. While our pipeline evaluation showed reasonable alignment with human judgments, these remain approximations of the underlying theoretical construct.

5 Evaluation

To evaluate IdeaBlocks and understand its role in supporting design exploration, we conducted two complementary studies. First, we conducted a within-subjects comparative study to validate whether IdeaBlocks effectively supports divergent intent expression and reuse, ultimately leading to more divergent design exploration than the baseline. This study addressed the following research questions:

- RQ1. How does IdeaBlocks support users in expressing and reusing their divergent intents?
- RQ2. How does the use of IdeaBlocks influence users' exploration process and patterns?
- RQ3. How does IdeaBlocks impact the ideational fluency and diversity of users' exploration?

We then conducted a three-day longitudinal study to more closely examine IdeaBlocks's intent-reuse features. While the formative study provided a foundation for the design of a divergent intent expression, it revealed broad possibilities for designing reuse features, leading to multiple intent-reusing features in IdeaBlocks. Therefore, we investigated how each feature is used in practice — when and why designers engage with it, and what purposes it serves in exploration. From these observations, we derived design implications for future intent-reusing tools. This study addressed the following research questions:

- RQ4. When and why do designers engage with reuse features with different units and mechanisms in practice?
- RQ5. How should future intent-reuse tools be designed to better support creative exploration?

6 Study 1: Comparative Study

We conducted a within-subjects comparative study with 12 participants to assess how well IdeaBlocks supports the expression and reuse of divergent intents. Additionally, we analyzed its impact on users' exploration patterns and the ideational fluency and diversity of their exploration outcomes to check if it helped divergent exploration.

6.1 Study Design

While divergent and convergent intents are interleaved in the design process, they serve different cognitive goals and are supported by different mechanisms. To examine whether and how IdeaBlocks's intent expression and reuse benefits divergent exploration, we intentionally focused more on the effects related to divergent intent in this study, rather than evaluating overall creative quality. We discussed the role of convergent processes in overall creative outcomes in Section 8.3.

6.1.1 Participants. We recruited 12 participants (5 females; M=25, SD=1.71) through an online recruitment post. All had a design

background through education or professional experience. Six were graduate students in design, three were final-year undergraduates in design, two held design degrees and were working, and one had no formal degree but worked as a graphic designer. Details are provided in Appendix C.2. The study lasted about 1.5 hours, and participants were compensated 50,000 KRW (approx $34 USD).

6.1.2 Study Procedure. Participants performed two rounds of design exploration tasks, each using either IdeaBlocks or the baseline system. The task involved generating diverse design concepts for one of two design topics: *"An illustration for a book cover of a detective novel set in a future city"* or *"A mascot for a summer night urban film festival."* These topics are selected as they do not have a typical visual representation and involve diverse design elements to explore, so that participants can explore divergently. The order of topics and systems was counterbalanced across participants.

Each round began with a 10-minute tutorial on the given system, during which participants practiced with a sample design topic to familiarize themselves with the interface. They then performed the exploration task for 20 minutes while thinking aloud about how their ideas developed. After each round, participants completed a post-task survey. Following both sessions, we conducted a 15-minute semi-structured interview. The full set of interview questions is in Appendix C.3.

6.1.3 Systems. We designed a baseline system as a generative AI–based design exploration tool with the same canvas-based interface as IdeaBlocks, but with only conventional mechanisms for intent expression and reuse. A screenshot is shown in Figure 9 in the Appendix. In the baseline, users enter a free-form text prompt describing their desired exploration, and the system generates four images using DALL·E 3, the same image generation model as IdeaBlocks. Like IdeaBlocks, the baseline supports non-linear exploration through branching on the canvas. However, its reuse mechanism simply copies the text prompt from the preceding block when a new block is added and linked, without additional structuring or contextual adaptation. To offset the absence of IdeaBlocks's LLM-based pipelines for intent expression and reuse (e.g., interpreting divergent intent or context-aware block copying), participants in the baseline condition were given access to ChatGPT[4]. Critically, the only difference between the two conditions was whether divergent intent was expressed through structured Exploration Blocks or a free-form text prompt, allowing us to isolate the effects of IdeaBlocks's intent-expression and reuse mechanisms.

We excluded the project-level reuse feature from IdeaBlocks in this study, as it cannot be meaningfully evaluated within a 20-minute session using a single exploration topic. Its use was examined in the longitudinal study (Section 7) instead.

6.1.4 Measures. We collected both qualitative and quantitative measures to assess how the system supported the divergence of early-stage design exploration. The post-task survey included 7-point Likert ratings on system support for expression and reuse of divergent intent, perceived efficiency and diversity of idea exploration, and satisfaction with exploration outcomes. Additionally, the survey also included the Creativity Support Index (CSI) [11] and NASA-TLX [27]. We also analyzed timestamped behavioral logs of actions (e.g., block creation, image generation, feature use) to derive quantitative measures, including the number and rate of generated images and the diversity of outputs. Finally, we analyzed the structural characteristics of the exploration session using linkography [23], capturing branching patterns, convergence, reuse, and connectivity between ideas. We used the Wilcoxon signed-rank test for all Likert-scale survey responses. For log data, we first ran a Shapiro-Wilk test for normality and applied either a paired t-test or a Wilcoxon test accordingly.

[4]https://chat.openai.com/

6.2 Results

Our findings indicate that IdeaBlocks significantly enhanced users' ability to express and reuse their divergent intent. It also supported more interconnected and structured patterns of idea exploration, resulting in greater ideational fluency and a broader diversity of concepts explored. We present our results in alignment with our three research questions (RQ1-3).

6.2.1 Support for Expressing and Reusing Divergent Intents. To answer RQ1, we examined survey responses and log data to evaluate how effectively participants could express and reuse their divergent intents.

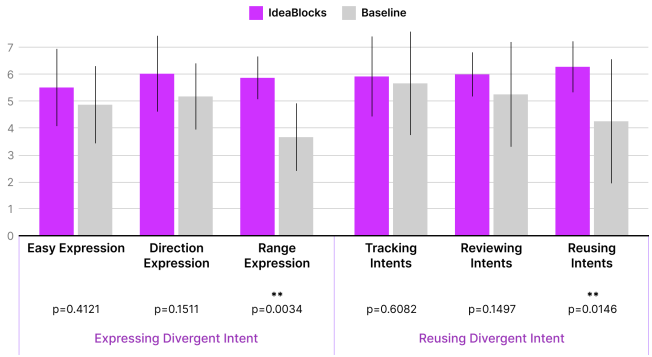


Figure 6: Survey responses on users' perceived support for intent expression (left) and intent reuse (right), with 95% confidence intervals. Participants rated IdeaBlocks significantly higher in their ability to express a range of intent and to reuse previously expressed intent.

Expressing Intent. As shown in Figure 6, participants rated IdeaBlocks significantly more helpful in expressing the *range* of their divergent intent (M=5.83, SD=0.80) compared to the baseline (M=3.67, SD=1.25 / p=0.0034, W=4.5). Participants largely attributed this advantage to the system's *typicality* slider, which allowed them to control how far the exploration should deviate, thereby better articulating the intended scope of their exploration. Although IdeaBlocks also received higher average scores in supporting the expression of *direction*, the difference was not statistically significant. This is likely because participants could still partially control directional intent through the baseline's text prompt.

This difference in perceived expressiveness was also reflected in participants' behaviors. Participants with IdeaBlocks created more input blocks (M=23.25, SD=7.97) compared to the baseline (M=13.17,

		IdeaBlocks		Baseline		Statistics	
		M	*SD*	*M*	*SD*	*p*	Sig.
Linkography Analysis	Avg. link distance	3.512	1.441	2.459	0.913	**0.0471**	*
	# of connected components	1.917	0.793	4.333	1.497	**0.0031**	**
	Link entropy	1.152	0.524	0.537	0.589	**0.0409**	*
Ideational Fluency	# of input block	23.25	7.97	13.17	2.66	**0.0010**	**
	# of generated images	123.42	41.66	58	13.27	**0.0005**	***
Diversity	Max cosine distance	0.6384	0.0505	0.5675	0.0897	**0.0457**	*

Table 1: Summary of quantitative results across RQ1–RQ3. Metrics include: (1) linkography-based structural measures (e.g., link distance, entropy), (2) behavioral indicators of ideational fluency (e.g., number of input blocks, generated images), and (3) diversity measures using CLIP-based cosine distance. (*: p < .050, **: p < .010, ***: p < .001)

SD=2.66 / *p*=0.0010, *t* = -4.41). During the interview, participants described how IdeaBlocks made intent expression easier. Rather than formulating a fully specified prompt upfront, they treated intent expression as an incremental and adjustable process. By separating property, direction, and range, IdeaBlocks allowed participants to externalize partial intent even when their goals were still unclear, and to refine it through interaction. As a result, participants perceived intent expression as a low-commitment activity, which encouraged more frequent articulation throughout the exploration session.

Reusing Intent. As illustrated in Figure 6, participants with IdeaBlocks reported greater effectiveness in reusing prior intents (*M*=6.25, *SD*=1.01) than the baseline (*M*=4.25, *SD*=2.31 / *p*=0.0146, *W*=2).

During the 20-minute sessions, participants using IdeaBlocks engaged in intent-reusing actions (copying blocks or paths) an average of 5.08 times (*SD*=4.36). Out of the 61 total instances of intent reuse, 41 targeted visual properties (e.g., image style, color palette). According to interview responses, participants found reuse particularly useful for such visual aspects, as these preferences tended to remain consistent across different ideas. In contrast, text-based properties (e.g., character entities) were seen as more context-dependent and were frequently changed between ideas. This contrast also appears in the longitudinal study findings in Section 7.2.1.

Ratings for tracking and reviewing intent did not differ significantly between conditions, as the interview results showed that these aspects were largely supported by the non-linear canvas interface that both baseline and IdeaBlocks shared. However, their reasons for doing so differed between conditions. P11 explained, *"In the baseline, I reviewed my exploration flow to assess the outputs and determine how to fix my next prompt. In IdeaBlocks, I do that to see which results were good to copy them directly."* This suggests that IdeaBlocks also shaped how participants reviewed prior work with a clearer purpose and actionable follow-up.

6.2.2 Exploration Patterns and Process. To address RQ2, we examined participants' exploration patterns using linkography [23], which analyzes the progression of ideas by representing each idea as a node and connections (e.g., references, derivations) as directed links. In our study, we constructed a linkograph for each session, where each node corresponds to an input block. Links captured (1) parent–child relationships formed through the non-linear canvas in both IdeaBlocks and the baseline, (2) intent-reuse actions unique to IdeaBlocks, and (3) explicit references to earlier ideas made during think-aloud. Example linkographs are provided in Figure 10 in the Appendix.

Diverse and Interconnected Idea Development. As shown in Table 1, IdeaBlocks sessions exhibited significantly higher link entropy (*M*=1.15, *SD*=0.52) than the baseline (*M*=0.54, *SD*=0.59 / *p*=0.0409), indicating that participants were more likely to pursue multiple parallel conceptual directions rather than a narrow or linear trajectory [23]. In addition, ideas generated with IdeaBlocks were connected across longer temporal and conceptual spans, as reflected by a greater average link distance (*M*=3.51, *SD*=1.44) compared to the baseline (*M*=2.46, *SD*=0.91 / *p*=0.0471). IdeaBlocks sessions also exhibited significantly fewer connected components (*M*=1.92, *SD*=0.79) than the baseline (*M*=4.33, *SD*=1.50 / *p*=0.0031), suggesting a more cohesive exploration structure overall.

Rather than reusing prior ideas to converge quickly or finalize results, participants described reusing earlier intents to keep diverse alternative directions available, which they referred to as "deferring commitment" to any single direction. In this sense, the increased connectivity and long-range linking observed with IdeaBlocks reflect deliberate maintenance of multiple viable directions, supporting sustained and exploratory navigation of the design space. Some participants also reflected on recognizing recurring exploration strategies through reuse, a pattern that we further examine in the longitudinal study in Section 7.

6.2.3 Ideational Fluency and Flexibility of Exploration. Creativity theory [24, 66] suggests that divergent thinking is driven by *ideational fluency* (the ability to generate many responses) and *flexibility* (the ability to explore diverse categories). Because IdeaBlocks targets early-stage expansion of the design space rather than refinement or evaluation, we focus on these two dimensions to assess whether it supports divergent exploration.

Ideational Fluency. As shown in Table 1, participants demonstrated significantly higher ideational fluency when using IdeaBlocks, generating more than twice as many images during the 20-minute session (*M*=123.42, *SD*=41.66) compared to the baseline (*M*=58.00, *SD*=13.27 / *p*=0.0005). Figure 8 shows that participants using IdeaBlocks also exhibited increasing image generation rates over time, whereas the baseline condition showed no such acceleration. The interview suggested that this increase was driven by their ability to build upon existing intentions rather than repeatedly starting from scratch. Reuse features allowed participants to

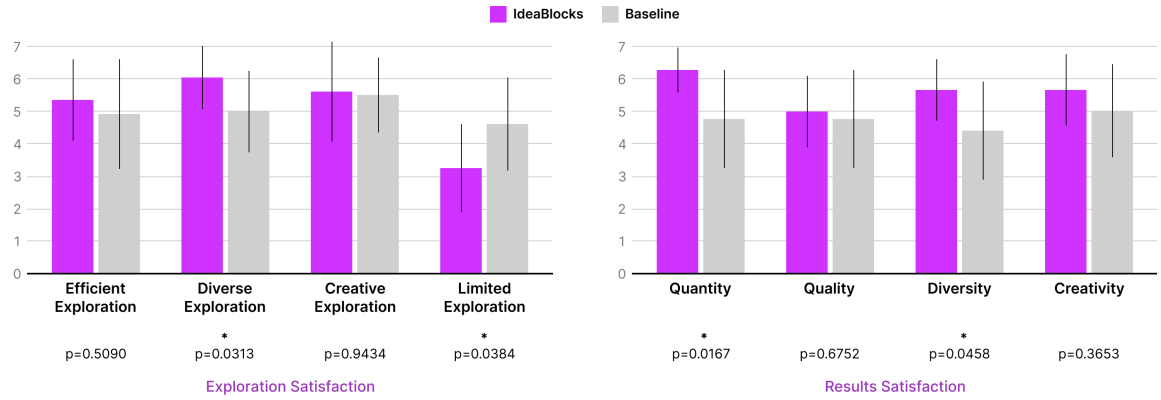


Figure 7: Survey responses on users' perceived satisfaction for the exploration process (top) and satisfaction with exploration results (right), with 95% confidence intervals. Participants rated IdeaBlocks significantly higher in supporting diverse exploration, reducing limited exploration (reversed item), and improving satisfaction with both the quantity and diversity of outcomes.

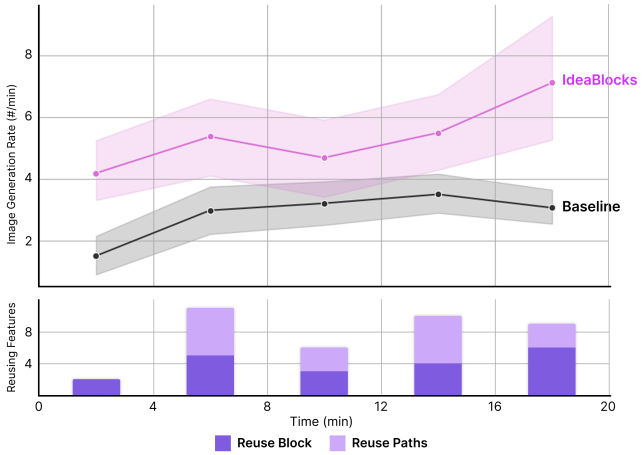


Figure 8: (Top) Image generation rate over time with 95% confidence intervals, grouped by 4-minute segments. IdeaBlocks users showed a steady increase, while the baseline plateaued mid-session. (Bottom) Frequency of reuse feature usage, split by block-level (*Copy Blocks*) and path-level (*Copy Paths*). The rising use of reuse features aligns with the acceleration in generation speed, suggesting their role in sustaining exploration momentum.

preserve previously articulated intent while exploring variations, reducing the effort required to explore new ideas. As a result, participants were able to sustain exploration momentum over time, which aligns with the observed acceleration in image generation during later stages of the session.

Participants reported significantly higher satisfaction with the quantity of ideas generated using IdeaBlocks (M=6.25, SD=0.72) compared to the baseline (M=4.75, SD=1.53 / p=0.016, W=2.5), but their ratings of exploration efficiency (M=5.33, SD=1.25) did not differ significantly from the baseline (M=4.92, SD=1.66; p=0.509, W=17). Interview responses suggest that participants interpreted efficiency differently: some associated it with how quickly they could converge on a final outcome (efficiency in convergence), while others focused on sustaining exploration (fluency in divergence). From this perspective, the observed increase in ideational fluency reflects a form of efficiency specific to divergent exploration, rather than faster convergence toward a single result.

Flexibility of the Results. Beyond the volume of ideas generated, flexibility captures whether exploration spans diverse conceptual directions rather than variations within a narrow scope. To assess this aspect, we analyzed the diversity of the generated outcomes at the image level. Using CLIP-based embeddings, we measured the maximum pairwise distance among images generated in each session as an indicator of visual diversity. As shown in Table 1, IdeaBlocks led to more visually diverse outcomes (M=0.6384, SD=0.0505) than the baseline (M=0.5675, SD=0.0897 / p=0.0457, t = -2.2529), suggesting that IdeaBlocks supported exploring a broader range of ideas within the given time. In the survey, participants also reported significantly higher satisfaction with the diversity of their outcomes when using IdeaBlocks (M=5.67, SD=0.94) than with baseline (M=4.42, SD=1.50 / p=0.043, W=6).

Perceived Creativity. Despite the positive results for divergent exploration, ratings of how much the system helped them explore creatively were similar (Baseline: M=5.50, SD=1.12 / IdeaBlocks: M=5.58, SD=1.55 / p=0.591, W=17.5), and satisfaction with the creativity of the final results did not differ significantly (Baseline: M=5.00, SD=1.47 / IdeaBlocks: M=5.67, SD=1.11 / p=0.278, W=15). Similarly, CSI scores showed no statistically significant differences between conditions. Together, these findings suggest that while IdeaBlocks effectively supports divergent exploration, the overall creativity requires complementary convergent processes as well, such as selection, refinement, and evaluation, which were intentionally held constant in this study. We discussed this complementary relationship between these two and suggested future directions in Section 8.3.

7 Study 2: Longitudinal Study

We conducted a three-day longitudinal study with six participants to understand when and how people use different intent-reuse

features, and how the system design could better support them. Based on the findings, we suggest design implications for future divergent intent reuse systems.

7.1 Study Design

7.1.1 Participants. We recruited 6 participants (3 females; M=25, SD=1.71) through an online recruitment post. Four were professional designers with prior work experience: two in graphic design (2 and 3 years), and two in UI design (6 months and 2 years). The other two were design students, one graduate and one undergraduate. Participant details are provided in Appendix C.2. One participant (D3) had also taken part in the comparative study (P9). Each participant was compensated 100,000 KRW (approx $72 USD).

7.1.2 Study Procedure. After the 30-minute system tutorial session, each participant used IdeaBlocks over three days, completing two 25-minute design exploration tasks per day (one mascot design and one book cover illustration task; ≈50 minutes of use per day). Throughout the study, IdeaBlocks collected usage logs, including timestamped actions such as the use of different intent-reuse features. After completing all tasks, we conducted a 30-minute semi-structured interview with each participant. The full interview protocol is provided in Appendix C.4.

7.2 Results

7.2.1 Block-level Reuse: Visual Assets as Designers' Palettes. During the study, participants reused single blocks, averaging 2.19 copy actions (SD=2.36) per 25-minute project. For each reuse, users could choose literal or adaptive pasting. Interestingly, participants strongly preferred literal reuse for visual properties such as image styles or color palettes (35 literal vs. 9 adaptive). Participants explained that these visual assets reflected their global preferences and were reused across projects. In contrast, for semantic properties such as characters' entities or poses, reuse was more evenly split (18 literal vs. 17 adaptive), suggesting that these aspects are more context-dependent and often require adaptation.

Several participants explicitly described block-level reuse as a way to build and reuse a "personal palette." D1 explained, *"Poses are highly dependent on context, so they are harder to reuse as-is, but image styles and color palettes are something I often bring back. They are closely tied to my personal taste."* D4 made an analogy to design tools, explaining that *"just like storing and reusing color swatches, I treated styles and colors as palettes to apply across projects."*

Even for non-visual semantic properties, participants preferred literal reuse when they had strong familiarity with a specific property type. For example, D5, who regularly designs YouTube thumbnails, explained: ' *'For background settings, I usually have a very clear sense of what I want, probably because I work with backgrounds a lot, so I already had plenty of ideas to reuse without any changes."*

> **Design Implication:** Systems should differentiate reuse mechanisms based on both property type and user expertise. Visual properties, or properties where users have strong familiarity, may benefit from literal reuse as cross-project palettes that reinforce identity and consistency. In contrast, semantic properties may be better supported through adaptive reuse, enabling greater contextual flexibility.

7.2.2 Path-level Reuse: Template-like Scaffolding. Participants also engaged with copying and pasting entire exploration paths, and we observed that this enabled users to reuse not just multiple blocks but the **structure of the exploration process** itself. They often described their exploration paths as recurring sequences of properties (e.g., entity → style → pose for D1) that functioned as reusable scaffolds. In interviews, they mentioned that, over multiple days of using IdeaBlocks, they began to recognize their own recurring strategies and increasingly relied on path reuse to reapply them quickly. This shift was reflected in the logs: use of path reuse grew steadily, from an average of 2.10 uses early on Day 1 to 2.90 on Day 2 and 3.30 on Day 3. Block- and project-level reuse showed no such upward trend.

D2 imagined formalizing such sequences into templates: *"If we can easily classify the blocks and make templates of the exploration order out of them, I think it would allow quick, adaptive ideation for many topics with just one click."*

> **Design Implication:** Systems should help users recognize and formalize their exploration strategies as path templates. These templates can serve as reusable scaffolds, allowing designers to quickly reapply their own strategies while adapting content to new contexts.

7.2.3 Project-level Reuse: Social Reuse as Collective Exploration. At the project level, IdeaBlocks allowed participants to import either their own previous projects or those created by others. We observed two distinct purposes for this: (1) *bootstrapping* and (2) *broadening options*.

For bootstrapping, participants reused the project at the beginning of exploration to avoid a blank start and quickly establish a foundation to build upon. In these cases, they tended to prefer smaller projects or their own prior project, and when reusing others' work, they selected projects whose intent was easy to understand so they could grasp the overall flow and extend it meaningfully. Because of this, participants valued IdeaBlocks's structured representation for making the original designer's reasoning legible. D1 reflected, *"At first, I resisted using someone else's project because it wasn't my flow of thought. But once I tried, I could infer their thought process and work on top of that."*

For broadening options, participants imported projects when they felt stuck and needed inspiration. In these cases, even when importing an entire project, participants treated it more like block- or path-level reuse — borrowing intermediate intent units or recurring strategies rather than the entire process. Interestingly, although IdeaBlocks provided adaptive reuse and recommendation features, participants overwhelmingly preferred exploring others' projects. D4 explained, *"Compared to generated options which could be random, others' projects felt more valuable because those people must have actually found good things during their exploration."*. This highlights how the combination of generative possibilities with human curation (embodied in others' projects) made them feel more trustworthy and valuable than purely algorithmic suggestions.

> **Design Implication:** Systems that support sharing and reusing others' exploration processes should treat the two purposes of reuse differently. For bootstrapping, systems should clearly represent the overall flow of intent evolution so users can build

on it. For broadening options, systems should foreground intermediate steps, keywords, and strategies instead of only full processes or the final outputs, to allow selective browsing.

7.2.4 Reuse and Creativity: Balancing Divergence and Convergence. Because intent reuse involves drawing on past actions, some participants expressed concern that it might steer exploration toward over-convergence.

At the same time, others highlighted how reuse could also spark more divergence and personal expression. Participants noted that the adaptive reuse options helped them sustain divergence even when reusing. D5 explained, *"Reusing elements didn't feel like convergent as when I copied the same one, I still saw other variations, and those adaptive options kept it fresh."* For literal reuse, several participants reframed it not as convergence but as a way to build one's own identity. D4 noted, *"Designers develop their own style. Saving and reusing that style is not a loss of originality — it's part of constructing identity."*

Participants also suggested system-level nudges to counter over-reliance on recent or popular items. D6 proposed, *"I could imagine the system blurs the recently reused blocks or paths so that older ones resurfaced, and it would encourage me to reuse in more diverse ways."* Extending this to social reuse, de-emphasizing overly popular projects or ideas could help avoid collective homogeneity and encourage discovery of less obvious alternatives.

Design Implication: Systems should offer adaptive variants alongside literal reuse to sustain divergence. Also, the system could introduce nudges such as recency fading to prevent over-reliance on the same items.

8 Discussion

8.1 Divergent Intent as a Parameterizable Construct

In IdeaBlocks, we conceptualized ***divergent intent*** not merely as a vague desire for alternatives, but as the explicit articulation of a parametric search space. While current generative AI models primarily optimize for convergent intent — refining output toward a specific, fixed target — early-stage design requires the flexibility to define and manipulate the boundaries of exploration. IdeaBlocks operationalizes this by decomposing intent into (property, direction, range), and providing a pipeline and interface to parametrically tune it. By shifting the interaction mechanism from instruction (telling the model "what" to generate) to sculpting (defining "where" and "how" to search), we transformed the generative model into a controllable engine for divergent thinking, allowing designers to structure the ill-defined design space.

While IdeaBlocks targets the visually open-ended nature of graphic design, its core mechanism of parameterizing divergent exploration into *Property*, *Direction*, and *Range* can be effectively adapted to other domains by redefining the specific parameters available for manipulation.

Across design tasks, the specific instantiation of these elements may vary. For example, while graphic design commonly foregrounds properties such as 'image style' or 'character entity', other domains may prioritize different aspects, such as 'layout' in UI/UX design or 'materials' in 3D product design. Similarly, the meaning and operationalization of *range* may vary across tasks. In IdeaBlocks, we instantiate range using *typicality* to capture how far exploration deviates from a familiar center. In other contexts, designers may instead wish to control parameters such as abstraction (literal to symbolic), visual fidelity (sketchy to polished), or constraint strength (loose to strict) to better align exploration with their own concerns. By externalizing divergent intent into explicit, task-appropriate parameters, future systems can adapt this approach to support exploratory needs across a wider range of creative practices.

8.2 Implications for Design: Scaffolding and Sustaining Divergence

IdeaBlocks demonstrates how modularizing and reusing divergent intent helps designers to explore better. In this section, we further discuss the future opportunities for intent-aware creativity support systems.

Regarding intent expression, our findings suggest a trade-off between explicit scaffolding and tacit fluency. IdeaBlocks lowers barriers for early-stage designers by externalizing the mental model of exploration (i.e., Property, Direction, Range). For novices who lack intuitive strategies to structure their search, the interface acts as a pedagogical scaffold to help them navigate the design space. However, this explicit parameterization may introduce friction for experts who rely on tacit intuition [51, 60]. To support expert users, future systems could be improved to infer users' tacit intent rather than requiring explicit input. For example, the system can capture tacit signals such as dwelling time on specific variations or spontaneous verbal reactions to automatically tune exploration parameters. This approach would allow the system to adaptively shape the search space without interrupting the designer's intuitive workflow.

Regarding intent reuse, participants from Study 2 noted the risk that reuse might lead to homogenization or over-convergence. While IdeaBlocks addresses this by offering adaptive reuse options, which mutate the original intent to fit new contexts, we suggest that future systems must proactively intervene to prevent the formation of this echo chamber. To mitigate reliance on familiar patterns, systems could introduce algorithmic nudges, for example, by temporarily de-emphasizing recently or frequently used blocks, allowing older or less popular strategies to resurface. By artificially diversifying the visibility of reusable intents, the system can disrupt potential fixation loops and encourage designers to adopt a diverse range of strategies.

8.3 The Diversity-Creativity Gap

IdeaBlocks was designed to isolate and strengthen the divergent phase of the creative process, enabling clearer examination of how such support impacts early-stage expansion of the design space. Within this scope, IdeaBlocks increased the breadth, structure, and continuity of exploration. However, these gains did not translate into significantly higher overall creativity ratings, suggesting that divergent exploration alone does not fully account for perceived creativity.

One explanation concerns how creative value is measured and experienced. Although IdeaBlocks accelerated the generation of options, this efficiency does not inherently lead to higher creativity.

Faster exploration can reduce the sense of authorship if transitions toward deliberate selection and refinement are not well supported, suggesting that objective diversity metrics capture only one dimension of creative work.

Another explanation relates to the role of convergent processes. While IdeaBlocks includes lightweight mechanisms that allow certain aspects of an exploration to be locked, these features do not constitute comprehensive support for convergent refinement. P8 stated, *"When I don't have a clear idea in mind, I like the broadness of system suggestions. But once I get a specific vision, it's difficult to control each block precisely and get the exact image that I want."* This tension between divergence and convergence is further reflected in the baseline condition: baseline participants may have reached a satisfactory result more quickly and stopped exploring, such that their lower ideational fluency reflects efficient convergence rather than a lack of divergent capability. Together, this suggests that creativity ratings may be shaped less by the volume of exploration than by how well the system supports the full arc from divergence to convergence.

From this perspective, future work can build on IdeaBlocks's divergent intent supports by enabling more fluid transitions between divergent and convergent intent. One promising direction is to allow exploratory outcomes (such as stabilized block or paths) to function as intermediate representations that inform subsequent convergent activities, either within the same system or through other tools. Supporting such connections iteratively would help preserve the value of divergent exploration as designers move toward more focused refinement.

9 Conclusion

This paper presents IdeaBlocks to support the divergent phase of design exploration by operationalizing divergent intent into reusable units comprised of property, direction, and range. By modularizing these intents, IdeaBlocks allows designers not only to shape the boundary of exploration but also to reuse the exploration logic across different contexts. Our evaluation showed that this structured approach significantly increases the diversity and connectivity of exploration. A longitudinal study revealed how designers appropriated reuse features in distinct ways — block reuse as palettes, path reuse as templates, and project reuse for bootstrapping or inspiration. These observations lead to design implications for future intent-reusing creativity support systems.

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References

[1] Krzysztof Adamkiewicz, Paweł Wojciech Woźniak, Julia Dominiak, Andrzej Romanowski, Jakob Karolus, and Stanislav Frolov. 2025. PromptMap: An Alternative Interaction Style for AI-Based Image Generation. In *Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25)*. Association for Computing Machinery, New York, NY, USA, 1162–1176. doi:10.1145/3708359.3712150

[2] Shm Garanganao Almeda, J.D. Zamfirescu-Pereira, Kyu Won Kim, Pradeep Mani Rathnam, and Bjoern Hartmann. 2024. Prompting for Discovery: Flexible Sense-Making for AI Art-Making with Dreamsheets. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 160, 17 pages. doi:10.1145/3613904.3642858

[3] Gabriela Goldschmidt and. 2016. Linkographic Evidence for Concurrent Divergent and Convergent Thinking in Creative Design. *Creativity Research Journal* 28, 2 (2016), 115–122. doi:10.1080/10400419.2016.1162497 arXiv:https://doi.org/10.1080/10400419.2016.1162497

[4] Tyler Angert, Miroslav Suzara, Jenny Han, Christopher Pondoc, and Hariharan Subramonyam. 2023. Spellburst: A Node-based Interface for Exploratory Creative Coding with Natural Language Prompts. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 100, 22 pages. doi:10.1145/3586183.3606719

[5] Michel Beaudouin-Lafon. 2000. Instrumental interaction: an interaction model for designing post-WIMP user interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (The Hague, The Netherlands) *(CHI '00)*. Association for Computing Machinery, New York, NY, USA, 446–453. doi:10.1145/332040.332473

[6] Stephen Brade, Bryan Wang, Mauricio Sousa, Sageev Oore, and Tovi Grossman. 2023. Promptify: Text-to-Image Generation through Interactive Prompt Exploration with Large Language Models. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 96, 14 pages. doi:10.1145/3586183.3606725

[7] Tim Brooks, Aleksander Holynski, and Alexei A. Efros. 2023. InstructPix2Pix: Learning to Follow Image Editing Instructions. arXiv:2211.09800 [cs.CV] https://arxiv.org/abs/2211.09800

[8] Bill Buxton. 2010. *Sketching user experiences: getting the design right and the right design.* Morgan kaufmann.

[9] Alice Cai, Steven R Rick, Jennifer L Heyman, Yanxia Zhang, Alexandre Filipowicz, Matthew Hong, Matt Klenk, and Thomas Malone. 2023. DesignAID: Using Generative AI and Semantic Diversity for Design Inspiration. In *Proceedings of The ACM Collective Intelligence Conference* (Delft, Netherlands) *(CI '23)*. Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3582269.3615596

[10] Tracy Cassidy. 2011. The Mood Board Process Modeled and Understood as a Qualitative Design Research Tool. *Fashion Practice* 3, 2 (2011), 225–251. doi:10.2752/175693811X13080607764854 arXiv:https://doi.org/10.2752/175693811X13080607764854

[11] Erin Cherry and Celine Latulipe. 2014. Quantifying the Creativity Support of Digital Tools through the Creativity Support Index. *ACM Trans. Comput.-Hum. Interact.* 21, 4, Article 21 (jun 2014), 25 pages. doi:10.1145/2617588

[12] DaEun Choi, Sumin Hong, Jeongeon Park, John Joon Young Chung, and Juho Kim. 2024. CreativeConnect: Supporting Reference Recombination for Graphic Design Ideation with Generative AI. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 1055, 25 pages. doi:10.1145/3613904.3642794

[13] DaEun Choi, Kihoon Son, Hyunjoon Jung, and Juho Kim. 2025. Expandora: Broadening Design Exploration with Text-to-Image Model. *arXiv preprint arXiv:2503.00791* (2025).

[14] John Joon Young Chung and Eytan Adar. 2023. Artinter: AI-Powered Boundary Objects for Commissioning Visual Arts. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference* (Pittsburgh, PA, USA) *(DIS '23)*. Association for Computing Machinery, New York, NY, USA, 1997–2018. doi:10.1145/3563657.3595961

[15] John Joon Young Chung and Max Kreminski. 2024. Patchview: LLM-powered Worldbuilding with Generative Dust and Magnet Visualization. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) *(UIST '24)*. Association for Computing Machinery, New York, NY, USA, Article 77, 19 pages. doi:10.1145/3654777.3676352

[16] Gregory Thomas Croisdale, John Joon Young Chung, Emily Huang, Gage Birchmeier, Xu Wang, and Anhong Guo. 2023. DeckFlow: A Card Game Interface for Exploring Generative Model Flows. In *Adjunct Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23 Adjunct)*. Association for Computing Machinery, New York, NY, USA, Article 89, 3 pages. doi:10.1145/3586182.3615821

[17] Kees Dorst and Nigel Cross. 2001. Creativity in the design process: co-evolution of problem–solution. *Design studies* 22, 5 (2001), 425–437.

[18] Steven P. Dow, Alana Glassco, Jonathan Kass, Melissa Schwarz, Daniel L. Schwartz, and Scott R. Klemmer. 2011. Parallel prototyping leads to better design results, more divergence, and increased self-efficacy. *ACM Trans. Comput.-Hum. Interact.* 17, 4, Article 18 (Dec. 2011), 24 pages. doi:10.1145/1879831.1879836

[19] Claudia Eckert and Martin Stacey. 2000. Sources of inspiration: a language of design. *Design studies* 21, 5 (2000), 523–538.

[20] Ronald A Finke, Thomas B Ward, and Steven M Smith. 1996. *Creative cognition: Theory, research, and applications*. MIT press.

[21] Leilei Gan, Zhiyang Teng, Yue Zhang, Linchao Zhu, Fei Wu, and Yi Yang. 2022. SemGloVe: Semantic Co-Occurrences for GloVe From BERT. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 30 (2022), 2696–2704. doi:10.1109/TASLP.2022.3197316

[22] Frederic Gmeiner, Nicolai Marquardt, Michael Bentley, Hugo Romat, Michel Pahud, David Brown, Asta Roseway, Nikolas Martelaro, Kenneth Holstein, Ken Hinckley, et al. 2025. Intent Tagging: Exploring Micro-Prompting Interactions for Supporting Granular Human-GenAI Co-Creation Workflows. arXiv:2502.18737 [cs.HC] https://arxiv.org/abs/2502.18737

[23] Gabriela Goldschmidt. 2014. *Linkography: unfolding the design process*. Mit Press.

[24] Joy Paul Guilford. 1967. The nature of human intelligence. (1967).

[25] Meera Hahn, Wenjun Zeng, Nithish Kannen, Rich Galt, Kartikeya Badola, Been Kim, and Zi Wang. 2024. Proactive Agents for Multi-Turn Text-to-Image Generation Under Uncertainty. arXiv:2412.06771 [cs.AI] https://arxiv.org/abs/2412.06771

[26] Evans Xu Han, Alice Qian Zhang, Haiyi Zhu, Hong Shen, Paul Pu Liang, and Jane Hsieh. 2025. POET: Supporting Prompting Creativity and Personalization with Automated Expansion of Text-to-Image Generation. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25)*. Association for Computing Machinery, New York, NY, USA, Article 162, 18 pages. doi:10.1145/3746059.3747710

[27] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Human Mental Workload*, Peter A. Hancock and Najmedin Meshkati (Eds.). Advances in Psychology, Vol. 52. North-Holland, 139–183. doi:10.1016/S0166-4115(08)62386-9

[28] Alexander Ivanov, David Ledo, Tovi Grossman, George Fitzmaurice, and Fraser Anderson. 2022. MoodCubes: Immersive Spaces for Collecting, Discovering and Envisioning Inspiration Materials. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference* (Virtual Event, Australia) *(DIS '22)*. Association for Computing Machinery, New York, NY, USA, 189–203. doi:10.1145/3532106.3533565

[29] Peiling Jiang, Jude Rayan, Steven P. Dow, and Haijun Xia. 2023. Graphologue: Exploring Large Language Model Responses with Interactive Diagrams. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 3, 20 pages. doi:10.1145/3586183.3606737

[30] Youwen Kang, Zhida Sun, Sitong Wang, Zeyu Huang, Ziming Wu, and Xiaojuan Ma. 2021. MetaMap: Supporting Visual Metaphor Ideation through Multi-dimensional Example-based Exploration. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) *(CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 427, 15 pages. doi:10.1145/3411764.3445325

[31] Joy Kim, Mira Dontcheva, Wilmot Li, Michael S. Bernstein, and Daniela Steinsapir. 2015. Motif: Supporting Novice Creativity through Expert Patterns. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) *(CHI '15)*. Association for Computing Machinery, New York, NY, USA, 1211–1220. doi:10.1145/2702123.2702507

[32] Kevin Gonyop Kim, Richard Lee Davis, Alessia Eletta Coppi, Alberto Cattaneo, and Pierre Dillenbourg. 2022. Mixplorer: Scaffolding Design Space Exploration through Genetic Recombination of Multiple Peoples' Designs to Support Novices' Creativity. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) *(CHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 308, 13 pages. doi:10.1145/3491102.3501854

[33] Tae Soo Kim, DaEun Choi, Yoonseo Choi, and Juho Kim. 2022. Stylette: Styling the Web with Natural Language. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) *(CHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 5, 17 pages. doi:10.1145/3491102.3501931

[34] Tae Soo Kim, Yoonjoo Lee, Minsuk Chang, and Juho Kim. 2023. Cells, Generators, and Lenses: Design Framework for Object-Oriented Interaction with Large Language Models. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 4, 18 pages. doi:10.1145/3586183.3606833

[35] Hyung-Kwon Ko, Gwanmo Park, Hyeon Jeon, Jaemin Jo, Juho Kim, and Jinwook Seo. 2023. Large-Scale Text-to-Image Generation Models for Visual Artists' Creative Works. In *Proceedings of the 28th International Conference on Intelligent User Interfaces* (Sydney, NSW, Australia) *(IUI '23)*. Association for Computing Machinery, New York, NY, USA, 919–933. doi:10.1145/3581641.3584078

[36] Janin Koch, Andrés Lucero, Lena Hegemann, and Antti Oulasvirta. 2019. May AI? Design Ideation with Cooperative Contextual Bandits. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) *(CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3290605.3300863

[37] Janin Koch, Nicolas Taffin, Andrés Lucero, and Wendy E. Mackay. 2020. SemanticCollage: Enriching Digital Mood Board Design with Semantic Labels. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (Eindhoven, Netherlands) *(DIS '20)*. Association for Computing Machinery, New York, NY, USA, 407–418. doi:10.1145/3357236.3395494

[38] Yuki Koyama and Masataka Goto. 2022. BO as Assistant: Using Bayesian Optimization for Asynchronously Generating Design Suggestions. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) *(UIST '22)*. Association for Computing Machinery, New York, NY, USA, Article 77, 14 pages. doi:10.1145/3526113.3545664

[39] Yuki Koyama, Daisuke Sakamoto, and Takeo Igarashi. 2016. SelPh: Progressive Learning and Support of Manual Photo Color Enhancement. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) *(CHI '16)*. Association for Computing Machinery, New York, NY, USA, 2520–2532. doi:10.1145/2858036.2858111

[40] Youn-Kyung Lim, Erik Stolterman, and Josh Tenenberg. 2008. The anatomy of prototypes: Prototypes as filters, prototypes as manifestations of design ideas. *ACM Trans. Comput.-Hum. Interact.* 15, 2, Article 7 (July 2008), 27 pages. doi:10.1145/1375761.1375762

[41] Vivian Liu, Han Qiao, and Lydia Chilton. 2022. Opal: Multimodal Image Generation for News Illustration. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) *(UIST '22)*. Association for Computing Machinery, New York, NY, USA, Article 73, 17 pages. doi:10.1145/3526113.3545621

[42] Andrés Lucero. 2015. Funky-Design-Spaces: Interactive Environments for Creativity Inspired by Observing Designers Making Mood Boards. In *Human-Computer Interaction – INTERACT 2015*, Julio Abascal, Simone Barbosa, Mirko Fetter, Tom Gross, Philippe Palanque, and Marco Winckler (Eds.). Springer International Publishing, Cham, 474–492.

[43] Wendy E. Mackay and Michel Beaudouin-Lafon. 2025. Interaction Substrates: Combining Power and Simplicity in Interactive Systems. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 687, 16 pages. doi:10.1145/3706598.3714006

[44] Atefeh Mahdavi Goloujeh, Anne Sullivan, and Brian Magerko. 2024. Is It AI or Is It Me? Understanding Users' Prompt Journey with Text-to-Image Generative AI Tools. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 183, 13 pages. doi:10.1145/3613904.3642861

[45] Justin Matejka, Michael Glueck, Erin Bradner, Ali Hashemi, Tovi Grossman, and George Fitzmaurice. 2018. Dream Lens: Exploration and Visualization of Large-Scale Generative Design Datasets. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) *(CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3173574.3173943

[46] Jasper O'Leary, Holger Winnemöller, Wilmot Li, Mira Dontcheva, and Morgan Dixon. 2018. Charrette: Supporting In-Person Discussions around Iterations in User Interface Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) *(CHI '18)*. Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3173574.3174109

[47] Xiaohan Peng, Janin Koch, and Wendy E. Mackay. 2024. DesignPrompt: Using Multimodal Interaction for Design Exploration with Generative AI. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) *(DIS '24)*. Association for Computing Machinery, New York, NY, USA, 804–818. doi:10.1145/3643834.3661588

[48] Yingzhe Peng, Xiaoting Qin, Zhiyang Zhang, Jue Zhang, Qingwei Lin, Xu Yang, Dongmei Zhang, Saravan Rajmohan, and Qi Zhang. 2025. Navigating the Unknown: A Chat-Based Collaborative Interface for Personalized Exploratory Tasks. In *Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25)*. Association for Computing Machinery, New York, NY, USA, 1048–1063. doi:10.1145/3708359.3712093

[49] Jeffrey Pennington, Richard Socher, and Christopher Manning. 2014. GloVe: Global Vectors for Word Representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Alessandro Moschitti, Bo Pang, and Walter Daelemans (Eds.). Association for Computational Linguistics, Doha, Qatar, 1532–1543. doi:10.3115/v1/D14-1162

[50] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. 2023. SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis. arXiv:2307.01952 [cs.CV] https://arxiv.org/abs/2307.01952

[51] Michael Polanyi. 2009. The tacit dimension. In *Knowledge in organisations*. Routledge, 135–146.

[52] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*, Marina Meila and Tong Zhang (Eds.). PMLR, 8748–8763. https://proceedings.mlr.press/v139/radford21a.html

[53] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. arXiv:1908.10084 [cs.CL] https://arxiv.org/abs/

1908.10084

[54] Nathalie Riche, Anna Offenwanger, Frederic Gmeiner, David Brown, Hugo Romat, Michel Pahud, Nicolai Marquardt, Kori Inkpen, and Ken Hinckley. 2025. AI-Instruments: Embodying Prompts as Instruments to Abstract & Reflect Graphical Interface Commands as General-Purpose Tools. arXiv:2502.18736 [cs.HC] https://arxiv.org/abs/2502.18736

[55] ELEANOR H. ROSCH. 1973. On the Internal Structure of Perceptual and Semantic Categories. In *Cognitive Development and Acquisition of Language*, Timothy E. Moore (Ed.). Academic Press, San Diego, 111–144. doi:10.1016/B978-0-12-505850-6.50010-4

[56] Donald A. Schon and Vincent DeSanctis. 1986. The Reflective Practitioner: How Professionals Think in Action. *The Journal of Continuing Higher Education* 34, 3 (1986), 29–30. doi:10.1080/07377366.1986.10401080 arXiv:https://doi.org/10.1080/07377366.1986.10401080

[57] Xinyu Shi, Yinghou Wang, Ryan Rossi, and Jian Zhao. 2025. Brickify: Enabling Expressive Design Intent Specification through Direct Manipulation on Design Tokens. arXiv:2502.21219 [cs.HC] https://arxiv.org/abs/2502.21219

[58] Ben Shneiderman. 2007. Creativity support tools: accelerating discovery and innovation. *Commun. ACM* 50, 12 (Dec. 2007), 20–32. doi:10.1145/1323688.1323689

[59] Herbert A Simon. 1996. The Sciences of the Artificial (3e éd.).

[60] Kihoon Son, DaEun Choi, Tae Soo Kim, and Juho Kim. 2024. Demystifying Tacit Knowledge in Graphic Design: Characteristics, Instances, Approaches, and Guidelines. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 221, 18 pages. doi:10.1145/3613904.3642886

[61] Kihoon Son, DaEun Choi, Tae Soo Kim, Young-Ho Kim, and Juho Kim. 2024. GenQuery: Supporting Expressive Visual Search with Generative Models. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 180, 19 pages. doi:10.1145/3613904.3642847

[62] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminate: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 644, 26 pages. doi:10.1145/3613904.3642400

[63] Sangho Suh, Bryan Min, Srishti Palani, and Haijun Xia. 2023. Sensecape: Enabling Multilevel Exploration and Sensemaking with Large Language Models. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 1, 18 pages. doi:10.1145/3586183.3606756

[64] Michael Terry and Elizabeth D. Mynatt. 2002. Recognizing creative needs in user interface design. In *Proceedings of the 4th Conference on Creativity & Cognition* (Loughborough, UK) *(C&C '02)*. Association for Computing Machinery, New York, NY, USA, 38–44. doi:10.1145/581710.581718

[65] Michael Terry, Elizabeth D. Mynatt, Kumiyo Nakakoji, and Yasuhiro Yamamoto. 2004. Variation in element and action: supporting simultaneous development of alternative solutions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vienna, Austria) *(CHI '04)*. Association for Computing Machinery, New York, NY, USA, 711–718. doi:10.1145/985692.985782

[66] E Paul Torrance. 1966. Torrance tests of creative thinking. *Educational and psychological measurement* (1966).

[67] Sitong Wang, Savvas Petridis, Taeahn Kwon, Xiaojuan Ma, and Lydia B Chilton. 2023. PopBlends: Strategies for Conceptual Blending with Large Language Models. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 435, 19 pages. doi:10.1145/3544548.3580948

[68] Yunlong Wang, Shuyuan Shen, and Brian Y Lim. 2023. RePrompt: Automatic Prompt Editing to Refine AI-Generative Art Towards Precise Expressions. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 22, 29 pages. doi:10.1145/3544548.3581402

[69] Zhijie Wang, Yuheng Huang, Da Song, Lei Ma, and Tianyi Zhang. 2024. PromptCharm: Text-to-Image Generation through Multi-modal Prompting and Refinement. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 185, 21 pages. doi:10.1145/3613904.3642803

[70] Xiaotong (Tone) Xu, Srishti Palani, Azzaya Munkhbat, Tiffany Lee, and Steven P. Dow. 2024. Idea-Centric Search: Four Patterns of Information Seeking During Creative Ideation. In *Proceedings of the 16th Conference on Creativity & Cognition* (Chicago, IL, USA) *(C&C '24)*. Association for Computing Machinery, New York, NY, USA, 280–291. doi:10.1145/3635636.3656193

[71] Zihan Yan, Chunxu Yang, Qihao Liang, and Xiang 'Anthony' Chen. 2023. XCreation: A Graph-based Crossmodal Generative Creativity Support Tool. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology* (San Francisco, CA, USA) *(UIST '23)*. Association for Computing Machinery, New York, NY, USA, Article 48, 15 pages. doi:10.1145/3586183.3606826

[72] Ryan Yen and Jian Zhao. 2024. Memolet: Reifying the Reuse of User-AI Conversational Memories. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) *(UIST '24)*. Association for Computing Machinery, New York, NY, USA, Article 58, 22 pages. doi:10.1145/3654777.3676388

[73] Loutfouz Zaman, Wolfgang Stuerzlinger, Christian Neugebauer, Rob Woodbury, Maher Elkhaldi, Naghmi Shireen, and Michael Terry. 2015. GEM-NI: A System for Creating and Managing Alternatives In Generative Design. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (Seoul, Republic of Korea) *(CHI '15)*. Association for Computing Machinery, New York, NY, USA, 1201–1210. doi:10.1145/2702123.2702398

[74] J.D. Zamfirescu-Pereira, Richmond Y. Wong, Bjoern Hartmann, and Qian Yang. 2023. Why Johnny Can't Prompt: How Non-AI Experts Try (and Fail) to Design LLM Prompts. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 437, 21 pages. doi:10.1145/3544548.3581388

[75] Jiayi Zhou, Renzhong Li, Junxiu Tang, Tan Tang, Haotian Li, Weiwei Cui, and Yingcai Wu. 2024. Understanding Nonlinear Collaboration between Human and AI Agents: A Co-design Framework for Creative Design. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 170, 16 pages. doi:10.1145/3613904.3642812

[76] Mingxu Zhou, Dengming Zhang, Weitao You, Ziqi Yu, Yifei Wu, Chenghao Pan, Huiting Liu, Tianyu Lao, and Pei Chen. 2024. StyleFactory: Towards Better Style Alignment in Image Creation through Style-Strength-Based Control and Evaluation. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology* (Pittsburgh, PA, USA) *(UIST '24)*. Association for Computing Machinery, New York, NY, USA, Article 107, 15 pages. doi:10.1145/3654777.3676370

A Formative Study

A.1 Details of Formative Study Participants

Details of the formative study participants are shown in Table 2. Participants responded to two 7-point Likert scale questions asking about their familiarity with generative AI ("How much do you understand how image generation AI models work?") in the enrollment survey.

PID	Age	Sex	Design Related Background	Frequency of Using Generative AI Model	Generative AI familiarity (7-scale)
P1	23	F	Undergrad student in design	Less than 1 time a month	3
P2	24	F	Master's student in design	Less than 1 time a month	3
P3	26	F	Master's student in design	Less than 1 time a month	3
P4	23	F	Worked as a freelancer designer	Never	2
P5	23	F	Undergrad student in design	Less than 1 time a month	3
P6	23	F	Bachelor's graduate in design	2-3 times a month	4
P7	32	M	Bachelor's graduate in design	1 time a week	4

Table 2: Participants' demographic information, design-related background, frequency of generative AI model usage, and self-reported familiarity on a 7-point Likert scale.

A.2 Thematic Analysis Codebook

This codebook outlines the three components of divergent intent that emerged from our inductive thematic analysis, with example quotes from the think-aloud sessions where participants articulated their divergent intent, the interaction experiences that contributed to each component, and friction points identified in interviews.

Theme	Definition	Representative Quotes	Interactions	Associated Pain Points
Property	The specific design attribute (e.g., Subject, Style, Composition) to focus on.	*"I like the character, just change the background style." (P7)* *"Keep the composition fixed, but try different rendering styles." (P4)*	Specifying which part of the prompt to modify, or through inpainting to isolate a specific design attribute	- Users cannot isolate specific properties e.g., changing the style prompt unintentionally alters the subject or composition
Direction	The qualitative vector or "vibe" the user wishes to steer towards.	*"I want to see something like this reference, but with a more 'sketchy' feel." (P5)* *"I'm looking for a 'cyberpunk' atmosphere, but less aggressive." (P2)*	Primarily through text prompting; occasionally through reference-based image-to-image generation	- Hard to verbalize the visual direction - When users want to update the direction after looking at the results, they struggle how to change them
Range	The magnitude of variance; the scope of exploration from the anchor.	*"I want to see totally different layout options, not just small tweaks." (P1)* *"Keep it mostly the same, just surprise me a little bit with the details." (P3)*	Image variation features and prompt modifiers (e.g., "various styles"), though these were often unreliable	- Users had hard time express how much the model deviates in a structured way - Reliance on trial-and-error or prompt hacking to generate the right level of diversity

Table 3: Thematic Mapping of Divergent Intent Components, Contributing Interaction Experiences, and Associated Pain Points.

B Pipeline Design

B.1 Image Generation Pipeline

To generate images that incorporate both previous exploration history and the user's new divergent intent, IdeaBlocks composes prompts using a two-step LLM pipeline. First, it prompts the LLM to extract detailed descriptions of previously explored properties from the most recent image in the connected block sequence. Then, it generates descriptions about the user's new exploration intent: for text-based properties, it uses the provided suggestion directly; for image-based properties, it prompts GPT-4o to produce a description of the new visual concept. GPT-4o then synthesizes these elements into a text-to-image prompt, which is passed to DALL-E 3 to generate the final image.

With the release of GPT-image-1, we simplified this pipeline using its image-editing model. The model takes the most recent image, its explored properties, and the user's new intent as input, blending the new property into the existing image while preserving previously established visual characteristics. The full prompts used in this pipeline are provided in the supplementary materials.

C Evaluation

C.1 Baseline System

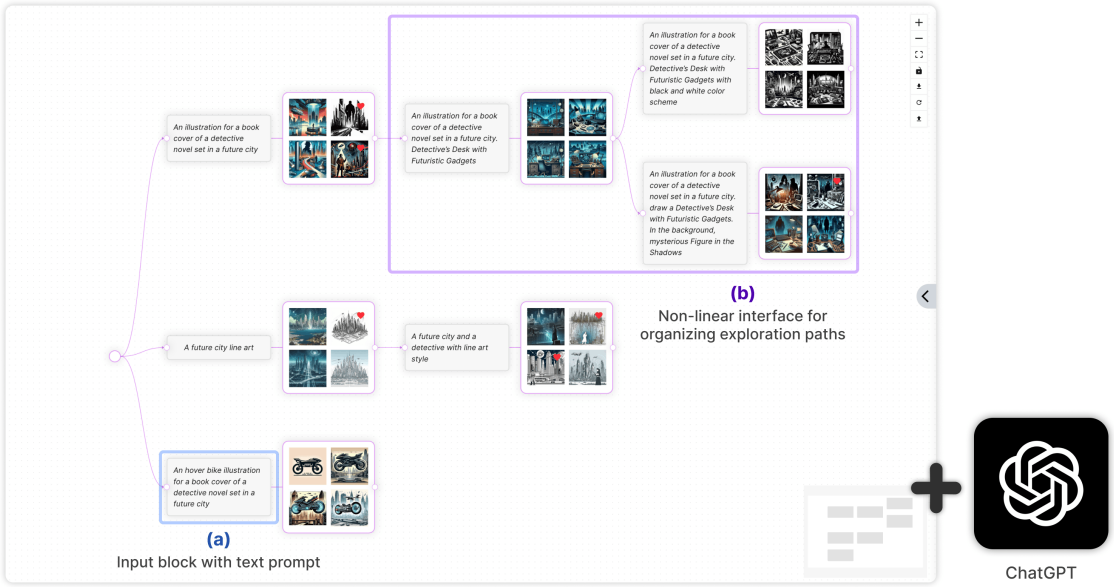


Figure 9: Screenshot of the baseline system. (a) Users initiate exploration by entering free-form text prompts. (b) The canvas-based interface supports non-linear branching of ideas, but without features for structured intent expression or reuse, such as direction/range inputs or path reuse. Users can use ChatGPT externally for assistance during exploration.

C.2 Details of Study Participants

C.3 Comparative Study Interview questions

These are the questions used for the semi-structured interview after the two design exploration sessions with baseline and IdeaBlocks.

IdeaBlocks vs. Baseline.

- What was the biggest difference when exploring design ideas using the two tools?
- Was there a difference in the way you explored design ideas when using the two tools?

Intent Expression.

- Was there a difference in how you conveyed your exploration intent when using the two tools? (e.g., text prompt vs. structured input)
- Which tool was better for fostering more creative/efficient exploration?

Revisiting & Intent Evolution.

- How did your exploration intent change when using the two tools?
- During the exploration process, did you continually refer to your previous results?
  - If so, how did your approach to referencing previous results differ when using each tool?

IdeaBlocks / Baseline Specific.

- What was the most useful feature of IdeaBlocks? In what cases was it particularly helpful?
- What was the least used feature of IdeaBlocks? Why did you use it less?
- In what cases and how did you use ChatGPT (in baseline condition)?

PID	Age	Sex	Design Related Background	Frequency of Using Generative AI Model	Familiarity of using GenAI	Understanding of GenAI
P1	24	F	Undergrad student in design	1–2 times a month	5	6
P2	22	M	Undergrad student in design	Once a week	5	5
P3	26	M	Ph.D. student in design	Once in 2-3 months	3	3
P4	28	M	Ph.D. student in design	1–2 times a month	4	5
P5	26	M	Ph.D. student in design	1–2 times a month	6	4
P6	26	M	Master's student in design	Once a week	4	6
P7	23	F	Master's student in design	2-5 times a week	4	3
P8	25	F	Bachelor's graduate in design	1–2 times a month	4	5
P9	25	M	Undergrad student in design*	Once a week	5	5
P10	24	F	Worked as a freelancer designer	1–2 times a month	5	2
P11	27	M	Master's graduate in design	2-5 times a week	6	4
P12	24	F	Master's student in design	Once in 2-3 months	4	4
D1	27	F	Ph.D. student in design	2-5 times a week	6	6
D2	25	F	Undergrad student in design	2-5 times a week	5	3
D3	25	M	UX/UI designer (2 years)*	2-5 times a week	5	4
D4	27	M	Game designer (3 years)	Once a week	6	5
D5	25	F	Freelancer graphic designer (2 years)	2-5 times a week	5	5
D6	19	M	UX/UI designer (6 months)	Almost every day	6	5

* P9 and D3 are the same participant, who transitioned from student to professional between the two studies.

Table 4: Participant details for the comparison study (P1–P12) and longitudinal study (D1–D6), including age, sex, design background, GenAI usage frequency, and self-reported familiarity and understanding of generative AI (7-point Likert scale).

Comparison with Usual Design Exploration Methods.

- Was the design exploration process using the provided tools similar to your usual method of exploring design ideas? If there were differences, please explain.
- The most significant difference might be the non-linear interface. How was your experience with that?

C.4 Longitudinal Study Interview questions

These are the questions used for the semi-structured interview after the two design exploration sessions with baseline and IdeaBlocks.

Overall Experience.

- How was your overall experience after using the system for three days?
- Compared to the way you usually explore design (e.g., Google Image Search, Pinterest, etc.), what felt most different?
- Compared to the way you usually use image generation models, what felt most different?

Experience with Intent Reusing Feature.

- Our system provided three ways to reuse previous explorations (Block, Path, Project). Which of the three did you use the most?
- In what situations did you use each unit of the reusing?
- When reusing a Project, did you ever start by referencing someone else's project? If so, what aspects did you usually reference? Was it easy to understand their exploration process?
- Each time you reused an exploration, our system provided two results (literal, adaptive). Which one did you prefer?
- Were the two approaches useful in different situations? If so, how did they differ?

Differences Depending on Context.

- Did your exploration approach change between the early and later stages of completing a task? If yes, how?
- From Day 1 to Day 3, did your way of using the system evolve over time? If yes, how?
- Did the familiarity of the task affect how you explored? If yes, how?
- Were there any other factors that influenced the way you used the system?

Future Improvement.

- Did you feel that the reuse feature supported creative exploration, or did it make the results converge instead?
- In what ways do you wish the system could better support your personal working style?

D Results

D.1 Example Linkographs from User Study Sessions

To further illustrate differences in exploration structure, Figure 10 shows qualitative examples of linkographs from participants' sessions in both conditions.

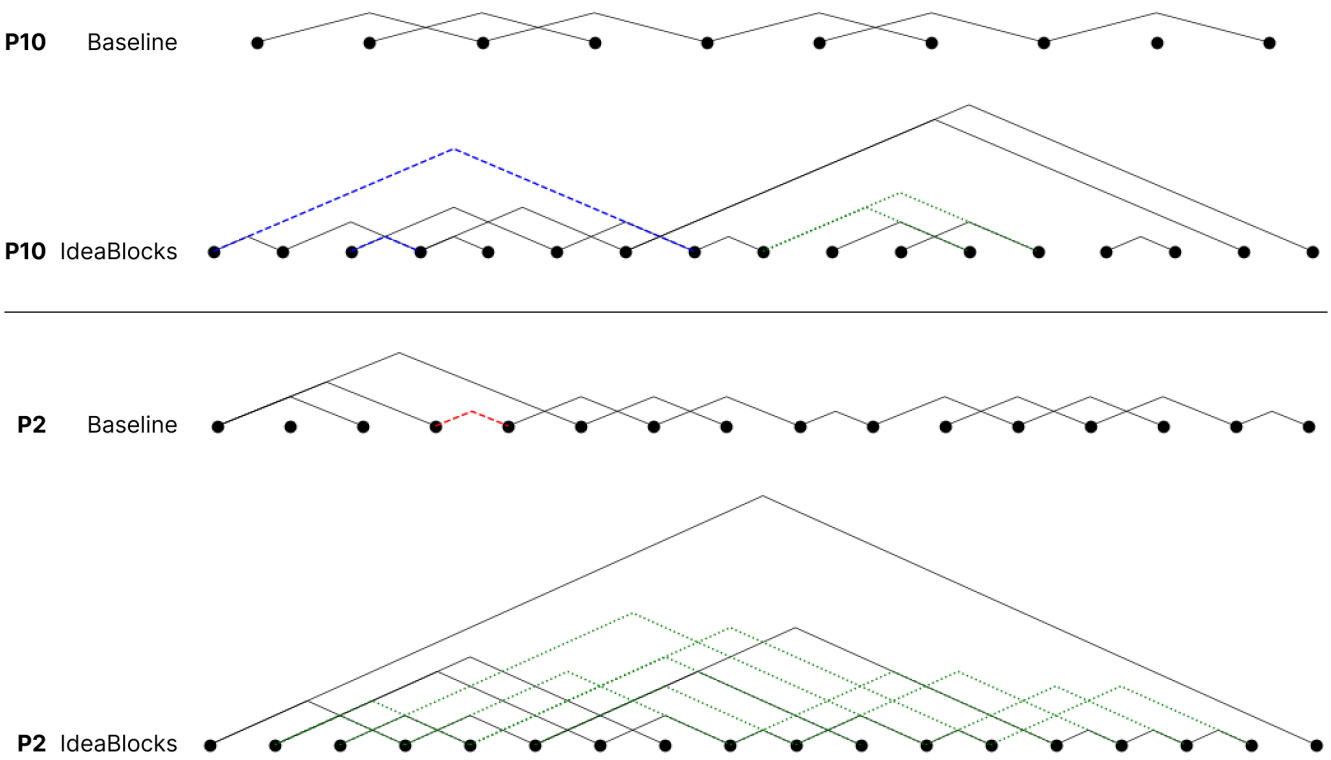


Figure 10: Example linkographs from two participants (P10 and P2) in both baseline and IdeaBlocks conditions. Each node represents an input block created during ideation. Solid black lines indicate parent-child connections, blue dashed lines indicate history-based reuse, green dotted lines indicate copy-based reuse, and red dotted lines indicate manually identified references based on think-aloud data. Compared to the baseline, participants using IdeaBlocks exhibited more frequent reuse and more interconnected idea structures.

D.2 CSI & NASA-TLX Survey Results

Table 5 shows the survey results for NASA-TLX and Creativity Support Index (CSI) between IdeaBlocks and baseline.

		IdeaBlocks		baseline		p
		M	SD	M	SD	
Creativity Support Index (CSI)	Enjoyment	6.17	0.85	5.38	1.49	0.1397
	Exploration	6.00	0.87	5.33	0.90	0.0901
	Expressiveness	5.79	0.75	5.00	1.26	0.0867
	Immersion	4.63	1.87	4.42	1.92	0.7993
	Results Worth Effort	5.21	1.31	5.00	1.10	0.6906
	Collaboration	5.63	0.77	5.17	1.57	0.3943
NASA-TLX	Mental demand	2.75	1.69	2.92	1.66	0.4314
	Physical demand	1.75	1.42	1.92	1.44	0.1573
	Temporal demand	2.08	1.19	2.25	1.42	0.7505
	Effort	2.25	1.01	2.75	1.42	0.3008
	Performance	5.08	1.19	4.67	1.43	0.4657
	Frustration	1.92	1.04	2.42	1.38	0.1573

Table 5: Survey results from NASA-TLX and Creativity Support Index (CSI) on a 7-point Likert scale. Each value shows the mean (M) and standard deviation (SD).